

The Labor Market Effects of Disability Hiring Quotas*

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Abstract

Over 115 countries use disability hiring quotas to address persistent underemployment among people with disabilities. This paper studies the labor market effects of enforcing such quotas in Brazil. Leveraging a reform that strengthened enforcement, I show that workers with disabilities in more exposed markets experienced higher employment and earnings. Using variation in inspections, I find that firms comply by hiring more workers with disabilities into low-wage jobs, who then experience slower wage growth. Outcomes for workers without disabilities and firms remain unchanged. Results indicate that modest quotas can raise net surplus by reducing discrimination at the hiring margin.

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1 Introduction

More than one billion people, around 16 percent of the global population, live with some form of disability (WHO, 2022). Yet, people with disabilities experience some of the largest labor market disadvantages: in most countries, their unemployment rates are at least twice that of the general population (UN, 2018). Because firms are often reluctant to hire people with disabilities, labor market policies that stimulate demand for these workers are as important as social insurance programs.¹ Implemented in more than 115 countries (see Figure 1), disability hiring quotas—whose origin dates back to World War I, when thousands of people were left disabled—are the most prevalent regulations aimed at reducing unemployment rates and the fiscal burden of social insurance (ILO, 2019).

This paper studies the labor market consequences of enforcing disability hiring quotas for both firms and workers. The merits of affirmative action hiring quotas for disadvantaged groups have long been debated. In competitive labor markets, classic models predict that such quotas may increase the earnings of disadvantaged workers but at the potential cost of displacing non-disadvantaged workers and reducing firm profits (Welch, 1976; Leonard, 1984; Griffin, 1992; Coate and Loury, 1993). Recent work highlighting imperfect competition and persistent discrimination suggests an alternative result. Firms may underemploy disadvantaged groups, in which case hiring quotas can increase employment and redistribute rents from firms to disadvantaged workers (Holzer and Neumark, 2000).

Despite the global prevalence of disability hiring quotas, comprehensive assessments of their labor market impacts remain quite challenging. At least two main reasons make investigating their impacts difficult. One reason is the data challenge: Few contexts link firms subject to quota regulations with workers with and without disabilities. Second, even where such data exist, these quotas are often weakly enforced, providing limited variation for causal identification. Recognizing these challenges, many governments have recently moved to strengthen quota enforcement (ILO, 2019).

This paper overcomes these challenges by exploiting several key features of the Brazilian context. First, Brazil has a disability hiring quota regulation and an enforcement mecha-

¹Non-discriminatory factors, such as work disincentives from social insurance, transportation barriers, lack of awareness about disability issues, and lower education levels, contribute to employers' reluctance to hire people with disabilities. However, experimental and observational evidence shows that employers remain less likely to hire individuals with disabilities even after accounting for these factors (Baldwin and Johnson, 1994; Ameri et al., 2018).

nism.² The regulation requires firms with at least 100 employees to allocate a minimum of 2 percent of positions to workers with disabilities. Labor inspections are the main enforcement tool: Firms found non-compliant during labor inspections receive a grace period to meet the mandated share or face heavy fines. These regulatory and enforcement structures, combined with high inspection frequency, create useful quasi-experimental variation. Second, detailed administrative data containing information on people with disabilities, including the type of disability, education, and occupation, provides a rare opportunity to address key data limitations. Third, Brazil’s large population and high disability prevalence provide statistical power when workers with disabilities are typically underrepresented in labor markets. Fourth, because disability insurance coverage is limited, access to social insurance programs is unlikely to be a major driver of disability-related employment gaps.

I begin my empirical analysis by quantifying the aggregate impacts of enforcing hiring quotas across local labor markets. Using Census data that cover the entire population, I exploit the introduction of a reform in the 2000s that established both the hiring quota regulation and labor inspections as its main enforcement mechanism. Along with the timing variation from the reform, I use geographic variation in the intensity of potential demand for workers with disabilities and enforcement of labor regulations. Specifically, I exploit the interaction between pre-reform firm size distribution and enforcement capacity as the source of cross-sectional differential exposure to the reform. The pre-reform size distribution predicts differential potential demand for people with disabilities induced by the regulation, while enforcement capacity is proxied by the driving distance to the nearest labor office.

Market-level estimates show that people with disabilities in more exposed local labor markets, such as São Paulo and Rio de Janeiro, experienced larger gains in formal employment and earnings, with formal sector employment rates about 10 percent higher a decade after the reform. I find no evidence of labor-labor substitution: Employment among people without disabilities and workers in the informal sector remains unchanged. While informative about aggregate effects, these market-level results provide limited insight into the mechanisms driving these changes at the firm level.

To understand how firms respond to quota enforcement, the second part of the paper

²The combination of quota regulations and enforcement mechanisms to promote the employment of people with disabilities is also adopted in other countries. Many European and Asian countries, such as France, Germany, Italy and South Korea, have analogous systems pairing hiring quotas with fines, inspections, or contribution schemes.

examines firm-level adjustments. The setting enables an event-study difference-in-differences design because, once inspected, firms may be evaluated for compliance with a wide range of labor regulations, but only firms with at least 100 employees can be assessed for compliance with disability hiring quotas. I compare the evolution of labor market outcomes for first-time inspected firms with pre-inspection employment just above and just below the 100-worker threshold. Because all these firms undergo the same inspection process, the post-inspection differences in outcomes can be attributed to adjustments to quota enforcement rather than to inspection itself. I show that firms on both sides of the threshold exhibit similar pre-inspection levels and trends in most outcomes, supporting the parallel trends assumption.

Using the universe of inspected firms linked to matched employer–employee data, I find that inspections lead to substantial increases in the employment of people with disabilities. On the extensive margin, firms subject to the quota are, on average, 8 percentage points more likely to hire at least one worker with a disability. On the intensive margin, these firms employ roughly 50 percent more workers with disabilities after inspection. Most new hires are concentrated in low-paying occupations and tend to have relatively mild disabilities, such as physical or hearing impairments. At the same time, there is no evidence of crowding out, as employment among workers without disabilities does not decline. Nor do firms become more likely to exit the formal sector or experience reductions in average wages, both of which serve as imperfect proxies for profits.

Turning to worker outcomes, the effects are concentrated among workers with disabilities. For this group, wage growth declines by roughly 6 percent at firms subject to the quota, whereas wage growth for comparable workers without disabilities is unaffected. These findings underscore important distributional consequences: Although the policy expands employment opportunities for the targeted group, it also increases within-firm wage disparities between workers with and without disabilities.

What can explain the lower wage growth for workers with disabilities? To interpret this pattern, I complement the administrative data with an original survey of human resources executives and personnel managers. I find evidence of two complementary frictions. First, imperfect competition: Workers with disabilities have more limited outside options and face greater mobility and search frictions, giving firms greater wage-setting power over this group. Consistent with this, wage effects are concentrated in more concentrated local labor markets. Second, dynamic statistical discrimination: Firms rely on noisy signals to infer workers' skills

for higher-level tasks and promotions, and may interpret signals from minority groups less accurately. Because quotas relax hiring standards, the average informativeness of signals from workers with disabilities declines. Accordingly, wage effects are concentrated among workers with disabilities with weaker opportunities to signal skills, such as those without a high school diploma or those in low-contact jobs. I also find no evidence supporting irreversible fixed costs, compensating differentials, or retaliation related to whistleblowing as alternative explanations for the decline in wage growth for workers with disabilities.

The final part of the paper evaluates the welfare consequences of quota enforcement. In my framework, the impact of disability quotas on firm profits also depends on the wedge between workers' marginal revenue products and their wages. Because direct estimates of the marginal revenue product of workers with disabilities are scarce, I develop a simple discrete choice framework that models firms' compliance decisions following inspections. I recover an implied marginal revenue product that still exceeds the wage, allowing me to reject that firms hire workers with disabilities at a loss. Combined with the absence of adverse impacts on firms and workers without disabilities, this finding suggests that compliance does not impose substantial costs on firms. At the same time, higher employment raises the economic surplus and fiscal contributions of people with disabilities. Taken together, these results indicate that enforcing modest disability hiring quotas generates net gains in aggregate welfare.

Conceptually, the combination of positive job surplus for newly hired workers with disabilities and no detectable adverse effects on firms or workers without disabilities is consistent with discrimination at the hiring margin as a key labor market friction limiting disability employment in the absence of hiring mandates. This interpretation aligns with my experimental survey. I document that Brazilian firms are 22 percent less likely to express interest in hiring a candidate with disabilities relative to an otherwise identical candidate without disabilities. In such labor markets, enforcing modest increases in mandated employment can reallocate job opportunities toward people with disabilities without reducing overall welfare.

This paper contributes to several strands of research. It relates to a large theoretical and empirical literature on the consequences of affirmative action hiring quotas in labor markets (Welch, 1976; Lundberg, 1991; Coate and Loury, 1993; Fang and Moro, 2011). Prior work has focused mainly on redistribution effects, documenting positive or neutral impacts on the employment of minority groups, including people with disabilities (Miller and Segal, 2012;

Lalive et al., 2013; Peck, 2017; Miller, 2017; Mori and Sakamoto, 2018; Prakash, 2020).³ My contribution is threefold. First, I provide new evidence on the aggregate effects of hiring quotas, leveraging unusually rich administrative data and quasi-experimental variation across local labor markets. These also allow me to track adjustments in both the formal and informal sectors. Second, relative to existing work, I provide more comprehensive estimates of the causal impact of hiring quotas on firm and worker outcomes, including wages, retention, and part-time employment. Third, I evaluate the welfare effects of quota enforcement and how these effects are distributed between firms and workers.

This work also relates to the literature on policies targeting people with disabilities. This literature has predominantly focused on supply-side interventions, analyzing how social insurance programs (e.g., disability insurance) and anti-discrimination regulations (e.g., the Americans with Disabilities Act) affect labor supply and consumption (DeLeire, 1997; Acemoglu and Angrist, 2001; Autor and Duggan, 2003; Maestas et al., 2013; Kostøl and Mogstad, 2014; Deshpande, 2016; Autor et al., 2019; Lise et al., 2023; Aizawa et al., 2024).⁴ In contrast, this paper examines the considerably less studied demand-side interventions (Lalive et al., 2013; Mori and Sakamoto, 2018; de Araújo et al., 2021; Palmer and Williams, 2023; Bloom et al., 2024), providing novel empirical evidence that regulatory enforcement and modest disability hiring mandates can operate as complements in raising firms' labor demand for people with disabilities, producing welfare gains by mitigating hiring-stage discrimination.⁵

At a broader level, this paper contributes to the literature on the incidence and efficiency implications of labor market regulations. In perfectly competitive labor markets, redistributive regulations cannot be justified on efficiency grounds. However, a growing body of evidence documents important frictions, such as employer discrimination (Glover et al., 2017; Benson et al., 2022) and imperfect competition, where firms exercise wage-setting power and

³Other studies examine whether affirmative action in education affects economic efficiency through changes in human capital or output (Bagde et al., 2016; Bleemer, 2022; Schaefer and Mankki, 2022). In leadership settings, mandated female representation on boards has been found to generate negative to no impacts on firm performance (Ahern and Dittmar, 2012; Matsa and Miller, 2013; Bertrand et al., 2019; Ferrari et al., 2021; Eckbo et al., 2022).

⁴A common policy among OECD countries is anti-discrimination regulation. The Americans with Disabilities Act, enacted in 1991 in the US, is one example. It prohibits discrimination based on disability and mandates reasonable accommodations for workers. Several studies indicate that the introduction of the ADA increased the costs of hiring individuals with work limitations, creating incentives for firms to avoid hiring them (DeLeire, 1997; Acemoglu and Angrist, 2001; Lise et al., 2023). Similar findings of limited employment effects have been documented in other contexts (Bell and Heitmueller, 2009; Lalive et al., 2013).

⁵Other studies also show that hiring quotas alone do not increase employment for people with disabilities in Brazil (Costilla et al., 2002; de Araújo et al., 2021; de Souza, 2023; Berlinski and Gagete-Miranda, 2025).

employ fewer workers than would be efficient (Card et al., 2013). In such labor markets characterized by frictions, regulations can potentially enhance efficiency while promoting redistribution (Manning, 2011). A few examples of price regulations studied in this literature include payroll taxes (Saez et al., 2019), minimum wages (Harasztosi and Lindner, 2019), wage floors (Card and Cardoso, 2021), and hiring credits (Cahuc et al., 2019). Wage subsidies represent another prominent price-based regulation aimed at expanding employment among disadvantaged groups (Weitzman, 1974). Yet, most evidence points to low take-up rates and limited effectiveness in increasing employment (Katz, 1996; Bartik, 2001; Huttunen et al., 2013; Jain et al., 2025).⁶ This paper instead studies a quantity-based regulation and shows that, with credible enforcement, modest disability quotas can increase employment and generate welfare gains, potentially serving as an alternative to price-based policies.

The remainder of this paper is structured as follows. Sections 2 and 3 describe the institutional context and data. Section 4 presents the aggregate impacts of enforcing the quota policy. Section 5 analyzes how firms make adjustments and potential mechanisms underlying some results. In Section 6, I discuss implications for welfare. Section 7 concludes.

2 Institutional Context

2.1 Disability Gaps in the Labor Market

In most countries, the employment gap between people with and without disabilities remains large, contributing to high rates of welfare dependency and poverty. Unemployment among working-age individuals with disabilities is at least twice the level of the general population (ILO, 2007; Mizunoya and Mitra, 2013; UN, 2018). Several non-discriminatory factors, such as work disincentives from social insurance, transportation barriers, limited awareness of disability issues, and lower educational attainment, partly explain these disparities. However, employers’ reluctance to hire workers with disabilities remains a major challenge. Observational studies find that firms are significantly less likely to employ people with disabilities (Baldwin and Johnson, 1994), and randomized audit experiments reveal lower callback rates for applicants with disabilities (Ameri et al., 2018; Bjørnshagen and Ugreninov, 2021; Bellemare et al., 2023). In Brazil, my experimental survey results show

⁶In contrast, wage subsidies have been shown to stabilize labor demand during recessions (Neumark and Grijalva, 2017; Schoefer, 2021).

that firms are 22 percent less likely to express interest in a candidate with a disability relative to an otherwise identical candidate without one. Concerns about workplace accommodation costs or productivity differences do not fully account for the observed gaps.

Until 2015, Brazil—like many countries, including Germany, Hungary, Poland, Sweden, and Turkey—adopted a predominantly medical approach to disability. Under this framework, disability was defined as an impairment of a physical, mental, intellectual, or sensory nature that, in interaction with various barriers, may hinder the full and effective participation of individuals in society on an equal basis with others. According to the 2010 Census, 17.2 percent of working-age Brazilians (aged 25–54) reported living with some form of disability, and 6.7 percent reported a severe disability.⁷ Despite the well-known challenges of defining and measuring disability, and the absence of fully standardized international statistics, these rates are comparable to those in other countries, including New Zealand, the United States, and the United Kingdom (Mitra and Yap, 2021).

Despite their high prevalence, people with disabilities remain vastly underrepresented in the labor market. Data from the 2010 Census reveal large employment gaps among the working-age population. As shown in Table A1, Appendix A, 78.8 percent of individuals without disabilities aged 25–54 are economically active. In contrast, those with some (severe) disabilities are 4.9 (19) percentage points less likely to participate in the labor force. Among economically active individuals, people with some (severe) disabilities face involuntary unemployment rates that are 4.6 (6.4) percentage points higher, indicating excess labor supply. Conditional on employment, workers with disabilities are 2.1 to 3.3 percentage points more likely to work in the informal sector and earn 12 to 18.2 percent lower wages than those without disabilities. These sizable gaps, unexplained by differences in location, education, potential experience, occupation, or sector, highlight substantial labor market disparities *even among people with disabilities who are able and willing to work*.

A means-tested unconditional cash transfer program, the *Benefício de Prestação Continuada*, provides a monthly benefit equivalent to one minimum wage (about USD 260 in 2023) to individuals with severe disabilities living in households with per capita income below one-quarter of the minimum wage. Eligibility does not require prior contributions to social security. On the other hand, coverage is highly restrictive, as only individuals with

⁷The Census asks whether respondents experience severe, some, or no difficulty in seeing, hearing, and walking or climbing stairs. It also records whether an individual has a permanent intellectual disability.

severe disabilities that prevent independent living and work capacity qualify for the transfer. In 2010, fewer than 4 percent of people with disabilities received the benefit, indicating that work constitutes the primary source of income for many individuals with disabilities.⁸

2.2 The Disability Reform

In line with global efforts to reduce disability inequality, Brazil enacted the National Policy for People with Disabilities (Decree 3,298) in 1999, a landmark piece of disability legislation that took effect in 2000 (hereafter the disability reform). Although a hiring quota for people with disabilities had been introduced in 1991 under Law 8,213, it remained largely ineffectual due to the absence of clear disability definitions and formal enforcement mechanisms. Decree 3,298 provided the first comprehensive legal framework supporting the employment of Brazilians with disabilities and has two key elements relevant to this paper.

First, the reform finally regulated the affirmative action disability hiring quota (henceforth AA quota), which requires firms in the formal private sector with at least 100 employees to allocate a minimum of 2 percent of positions with people with disabilities or beneficiaries of vocational rehabilitation under Social Security. The required share increases with firm size: firms with 201–500 employees must fill 3 percent of positions, those with 501–1,000 employees 4 percent, and firms with more than 1,000 employees 5 percent. The reform also established a legal definition of disability and provided a preliminary list of qualifying medical conditions, including mobility and physical impairments (e.g., paraplegia, monoplegia, limb amputation or absence, and congenital or acquired deformities), moderate to profound hearing loss, moderate visual impairment to blindness, and certain mental disorders. A complete list of qualifying disabilities is provided in Appendix A.2.

Second, the reform designated general labor inspections as the primary mechanism for enforcing the AA quota. Due to regulatory complexities, systematic monitoring only became feasible after 2003, when the Ministry of Labor began collecting information on workers with disabilities in the formal sector and introduced administrative fines for non-compliance (Thomasi et al., 2018).⁹ Appendix A.3 describes this regulatory process in detail.

⁸In contrast to an estimated 45.6 million individuals reporting some form of disability, only 1.7 million received the means-tested unconditional cash transfer in August 2010.

⁹Normative Instruction SIT 20 (2001) defined inspection procedures, and *Portaria* 1,199 (2003) established penalties for non-compliance.

2.3 Enforcement of Disability Hiring Quotas

The labor market in Brazil has a rigid legal structure with numerous mandated benefits to workers (e.g., minimum wage, unemployment insurance, and severance pay), high dismissal costs, and burdensome tax requirements (Haanwinckel and Soares, 2021). Imperfect compliance with labor regulations is quite common. In many cases, firms in the formal sector hire workers without fully complying with several regulations, such as payments to social security and disability quotas, or even hire informal workers “off the books” (Ulyssea, 2018).

The Ministry of Labor verifies compliance with many dimensions of labor regulations and uses general labor inspections as the main enforcement tool. The enforcement capacity has a decentralized structure: Each state has its own state labor office (*delegacia*) located in the state capital, and each state may also have additional local labor offices (*subdelegacias*) in other municipalities, depending on the state’s size and economic importance.¹⁰ Inspections can be triggered by random firm audits or anonymous reports from numerous sources, such as workers, labor unions, or the prosecutor’s office (Almeida and Carneiro, 2012).

The technology of enforcement is straightforward. Inspectors are assigned to a specific local labor office and must travel by car to inspect firms. This generates substantial spatial variation since smaller distances to the labor office imply a higher probability of inspection and, as a result, enforcement capacity (Ponczek and Ulyssea, 2021). Once inspected, firms may be assessed for compliance with many dimensions of labor regulations, including formal registration, severance and overtime pay, minimum wage, and number of hours worked. The key feature, fundamental to my research design, is that *only* firms with at least 100 full-time employees may also be required to fill 2 percent of positions with workers with disabilities.

Firms that fail to meet the AA quota may be notified and granted a grace period to comply with the regulation after an initial inspection. In case of continued non-compliance, a heavy fine, which typically ranges from 2.35 to 235 monthly minimum wages per missing worker, may be imposed unless the firm shows concrete evidence of effort and failure in hiring workers with disabilities.¹¹ To qualify for the job reserved to the AA quota, the worker

¹⁰Brazil has 27 states, implying that there are 27 state labor offices (*delegacia*). States with low population density, such as Acre, Amapá, and Tocantins, have only one labor office (*delegacias*). More dynamic states, such as São Paulo and Minas Gerais, have at least 20 local labor offices (*subdelegacias*) besides their own state labor office (*delegacia*).

¹¹The final amount of fines is a function of the number of workers with disabilities or rehabilitated workers that firms fail to hire in compliance with the AA quota regulation. It may also depend on numerous factors, such as firm size, recidivism, economic sector, geographic location, and local labor office. In some cases,

needs to have a medical report containing the type of disability and its code following the International Classification of Diseases (ICD). To minimize moral hazard or fraud in the hiring process, the worker must provide consent to disclose this information. The penalties for providing fraudulent information can be quite severe.¹²

In August 2012, the Ministry of Labor introduced Normative Instruction SIT 98, which, among other provisions, created the possibility of targeted inspections specifically focusing non-compliant firms (de Souza, 2023; Berlinski and Gagete-Miranda, 2025). Because my administrative data do not distinguish general inspections from these targeted inspections, and this distinction is central to the empirical strategy described in Section 5, my analysis is restricted to firms inspected before this regulatory change.¹³

3 Data

This paper uses four primary data sources: Data on labor inspections provided by the Ministry of Labor, the matched employer-employee records covering the entire formal sector, the decennial Demographic Census data covering the Brazilian population, and a small-scale original survey I conducted on firms’ perceptions of workers with disabilities.

Labor Inspections. The first source of data summarizes labor inspections conducted by the Ministry of Labor. The data contain detailed information on tax identifiers to be matched with the employee-employer records, the dates on which the reports are submitted (used here

firms may be exempted from the hiring quota if they present proof of attempts to comply with AA quota regulation (e.g., frequent job postings targeting people with disabilities, lack of qualified candidates, etc.). In these cases, the firms may be granted an extended grace period and asked to sign a statement confirming their commitment to comply with the AA quota regulation. In addition, several firms, such as ones from civil construction and oil and gas industries, can obtain an exemption from the AA quota via lawsuits alleging it is not possible to guarantee safe work conditions to workers with disabilities.

¹²Informal conversations with inspectors indicate that, while isolated frauds can occur, systematic fraudulent schemes do not represent a major concern in Brazil because potential criminal, financial, and reputational consequences of fraud are significantly more severe than the administrative penalties for non-compliance.

¹³Complementary to this setting, recent works have examined other aspects of Brazil’s quota enforcement system. For example, Berlinski and Gagete-Miranda (2025) study enforcement spillovers across firm networks following regulatory changes implemented after 2012, while de Souza (2023) exploits targeted inspections triggered by quota non-compliance in very large firms, carried out by the National Coordination Office for the Promotion of Equal Opportunities and the Elimination of Discrimination at Work. By contrast, my analysis leverages the 2000 disability reform that regulated both the quota and its enforcement mechanism and focuses on first-time general, non-targeted inspections across firms near to the statutory 100-employee threshold. This distinction is important because first-time general inspections render inspection timing plausibly exogenous to firms’ prior quota compliance status, allowing me to isolate the causal effects of quota enforcement from broader enforcement dynamics. Sections 4 and 5 further discuss these details.

as the start dates of labor inspections), and the labor violations (e.g., formal registration, hours of work, severance pay, minimum wage, etc.) uncovered during labor inspections.

Firm- and Worker-Level Data. Labor market information is extracted from RAIS (*Relação Anual de Informações Sociais*), a matched employee-employer data from the Ministry of Labor. It provides a comprehensive and high-quality annual overview of the formal sector (Szerman, 2023). Firms report information on their workers, including hiring and separation dates, average annual wages, number of hours contracted, occupation, and demographic characteristics, including gender, age, educational level, race, disability status, and type of disability. Information on whether the worker has a disability and the type of disability started to be reported by firms in 2003.¹⁴

Census Data. The third dataset comes from the decennial Brazilian Demographic Census (2000 and 2010) collected by the Brazilian Institute of Geography and Statistics (IBGE). The Census provides detailed information on individuals' demographic and socioeconomic characteristics and labor market outcomes, including self-reported disability status and type, employment status, and income from both work and non-work sources (e.g., welfare benefits). These two waves allow me to construct variables of interest before and after the 2000 reform.

Compared with the matched employer-employee data, the Census has the advantage of capturing labor market outcomes outside the formal sector. Due to major methodological changes in disability-related questions between the 1991 and 2000 Censuses, I exclude data from 1991, the first Census to include disability questions. Detailed descriptions of these methodological changes are provided in Appendix A.4. Because oversight and monitoring of the affirmative action quota through labor inspections became feasible only after 2003, I treat 2000 as the pre-reform period. I define as persons with disabilities those individuals who report living with *some* or *severe* difficulty in at least one functional domain.

HR Survey. To better understand how people with disabilities are perceived in the labor market and to explore mechanisms underlying the empirical results, I designed and imple-

¹⁴Disability status and type of disability are rarely reported in the matched employee-employer records, limiting progress in research on the role of employers and workplace policies on disability disadvantage in labor markets. This limitation has been described in Baldwin and Johnson (2006): “there is also a serious shortage of data that support empirical analyses of the labor market experiences of persons with disabilities. (...) The ideal data set includes detailed information on employment, wages, work experience, and job characteristics, as well as information on health conditions, functional limitations, and disability status.”

mented a small-scale original survey of human resources executives and personnel managers. The survey features vignettes in which information about disability is experimentally varied, along with questions on attitudes toward affirmative action, and perceived challenges and constraints in hiring workers with disabilities. Appendix G provides a detailed description of the survey design and implementation.

Other Data. I rely on additional minor sources of data to conduct the main analyses. First, information on municipal characteristics, such as gross domestic product (GDP) and population, is extracted from Ipeadata. Second, I gather labor offices’ addresses from official sources and link them to their date of creation extracted from Ponczek and Ulyssea (2021) to maintain labor offices created before the enactment of the regulation. Third, I also use this information to construct a measure of enforcement capacity by computing the minimum distance between each micro-region and the nearest labor office using the Google Maps API.¹⁵ Fourth, I obtain data on administrative fines from the Ministry of Labor.

4 Aggregate Analysis

In this section, I analyze the aggregate effects of the disability reform on employment and wages for people with and without disabilities across local labor markets. The next section closely examines how firms respond to the AA quota scheme. The institutional context suggests that labor markets with a high concentration of large firms and frequent labor inspections are more likely to experience an increase in the demand for workers with disabilities after the disability reform. To gauge the aggregate effects of the reform on employment and earnings for people with and without disabilities across local labor markets, this section proposes an identification strategy that exploits two dimensions of the reform: the timing of its enactment and the spatial heterogeneity in potential exposure.

¹⁵I obtain the distance between the centroid of each municipality and the nearest labor office created before the 2000s. I then define the minimum distance of the municipalities that belong to each micro-region as the minimum distance between each micro-region and the nearest labor office, which is used as a proxy for enforcement capacity.

4.1 Exposure to the Disability Reform

I construct a measure of local exposure to the disability hiring reform in three steps. The goal is to capture pre-reform spatial heterogeneity in how binding the AA quota regulation would be across local labor markets and how likely it was to be enforced. First, I define local labor markets at the micro-region level. In Brazil, micro-regions aggregate economically integrated contiguous municipalities with similar economic and geographic characteristics, analogous to US commuting zones.

Second, I define potential demand for workers with disabilities across local labor markets as the share of people with disabilities that would potentially benefit from jobs reserved by the AA quota regulation:

$$potential\ demand_r^{pre} = \frac{potential\ jobs_r^{pre}}{total\ PwD_r^{pre}}. \quad (1)$$

The numerator, $potential\ jobs_r^{pre}$, represents the total number of jobs in the private sector available to people with disabilities in each micro-region r in case of perfect compliance with the AA quota. I calculate this number from the distribution of firm size in 1998, before the reform, extracted from RAIS data. The denominator, $total\ PwD_r^{pre}$, normalizes potential jobs by the pre-reform number of people with disabilities in each micro-region r , calculated from the 2000 Census. The term $potential\ demand_r^{pre}$ reflects how binding the AA quota regulation could be in each area and has a median ratio of 5 percent. I then classify micro-regions above and below this median as micro-regions with strong and weak potential demand. Figure B1, Appendix B, shows how potential demand varies geographically.

The third step consists of building off a measure of enforcement capacity developed by Almeida and Carneiro (2012) and used in other papers (Haanwinckel and Soares, 2021; Ponczek and Ulyssea, 2021) to exploit spatial heterogeneity in the enforcement of labor regulation. The Ministry of Labor adopts a straightforward technology of enforcement, requiring only two inputs: inspectors who are assigned to labor offices and travel distance by car between labor offices and inspected firms. In principle, micro-regions located farther away from labor offices are less likely to receive inspections. Figure B4, Appendix B, confirms this negative relationship between the number of inspections per firm and the distance to the nearest labor office. It motivates the use of minimum travel distance to the nearest labor office within each micro-region as a proxy for enforcement capacity. To ensure that

the measure of enforcement capacity does not respond to changes in local labor market conditions and adjustments to the AA quota regulation, I limit the analysis to labor offices created before the reform. The median distance is about 60 kilometers (or 66 minutes). I then classify micro-regions with pre-determined distances below and above the median as those with weak and strong enforcement capacity. Figure B2, Appendix B, illustrates the variation in enforcement level across micro-regions.

Finally, I interact potential demand and enforcement capacity to retrieve the cross-sectional *initial exposure* to the disability reform. Figure 2 illustrates the resulting four exposure categories. Intuitively, people with disabilities in areas with both strong potential demand and strong enforcement should experience the largest effects of the reform. Because both components are constructed using pre-reform data, the exposure measure is predetermined with respect to the introduction of the reform.

4.2 Empirical Strategy

My empirical strategy for the market-level analysis considers both the timing of the reform and the spatial heterogeneity in exposure to it. Put differently, I estimate the following regression model using the Census data:¹⁶

$$y_{rst} = \alpha + (\mathbf{1}_r^{SD,SE} \times Reform_t)\beta_1 + (\mathbf{1}_r^{SD,WE} \times Reform_t)\beta_2 + (\mathbf{1}_r^{WD,SE} \times Reform_t)\beta_3 + \alpha_r + \alpha_t + \alpha_s \times t + X_{r,2000}\lambda + \varepsilon_{rst}, \quad (2)$$

in which subscripts r , s , and t stand for micro-region, state, and time; the indicator variables $\mathbf{1}_r^{SD,SE}$, $\mathbf{1}_r^{SD,WE}$, and $\mathbf{1}_r^{WD,SE}$ represent micro-regions with strong potential demand and enforcement capacity, with strong potential demand and weak enforcement capacity, and with weak potential demand and strong enforcement capacity, respectively; $Reform_t$ is an indicator for the period after the disability reform; α_r and α_t delineate micro-region and time fixed effects; $\alpha_s \times t$ represents state-specific trends; $X_{r,2000}$ is a vector of baseline micro-region characteristics from 2000, such as gender composition, education levels, urbanization, unemployment, literacy, and GDP per capita, each interacted with time fixed effects. The

¹⁶The aggregate analysis relies on the Census for two reasons. First, it captures outcomes outside the formal sector. Second, information on disability status in the RAIS data is available only after 2003, following the reform. Section 4.3 presents a robustness exercise using RAIS data.

dependent variable, y_{rst} , represents the labor market outcome of interest. Standard errors are clustered at the micro-region level.

The coefficients of interest, β_1 to β_3 , capture differential labor market responses across micro-regions with varying levels of exposure, relative to those with both weak potential demand and weak enforcement capacity. To mitigate potential confounding from concurrent national or regional developments, the specification includes a rich set of fixed effects and trends. Micro-region fixed effects absorb time-invariant local characteristics, while year fixed effects capture nationwide shocks such as macroeconomic growth, educational expansion, and the rollout of social programs like *Bolsa Família*. In addition, state-specific time trends flexibly account for unobserved state-level dynamics, including differential economic growth or policy initiatives that may have coincided with the reform period.

Although the exposure measure is predetermined with respect to the introduction of the reform, it could still correlate with initial local labor market conditions for people with disabilities. For instance, regions with stronger pre-reform economic performance may have followed different trajectories afterward. To address this concern, I interact the vector of baseline controls, $X_{r,2000}$, with time effects, allowing for heterogeneous trends across regions with distinct initial characteristics.

The identifying assumption is that, conditional on these controls, the cross-sectional exposure measure is orthogonal to unobserved factors that could differentially affect labor market outcomes for people with disabilities after the reform. While this assumption cannot be tested directly, several pieces of evidence support it. First, the exposure measure is constructed entirely from pre-reform data. Second, although the Census data do not allow a formal pre-trend test, as 2000 is the first wave with reliable information on disability, this limitation is unlikely to invalidate identification. Complementary evidence from the RAIS data shows no differential pre-trends in labor market outcomes for people with disabilities across exposure groups (Section 4.3). Third, I assess effects on the informal sector and on workers without disabilities as placebo tests. While some unobserved characteristics correlated with the exposure measure could, in principle, remain unaccounted for, it is difficult to think of factors that would increase formal employment specifically among people with disabilities while leaving workers without disabilities and the informal sector unaffected by the reform. The lack of significant effects on these placebo groups provides reassurance that the results are not driven by confounding regional shocks. Importantly, the identification strat-

egy exploits pre-determined cross-sectional variation in exposure, with identification coming from differences in treatment intensity across regions, rather than from parallel pre-trends in the Census data.

Sample and Summary Statistics. Using individual-level data from the Census, I restrict the sample to the working-age population between 25 and 54 years old to focus on individuals with strong attachment to the labor force. For each micro-region, I compute employment and informality rates and average income separately for people with and without disabilities. Table C1, Appendix C, reports summary statistics for the variables used in the aggregate analysis and confirms that people with disabilities exhibit substantially worse labor market outcomes than those without disabilities. Most demographic, economic, and labor market indicators display considerable dispersion across micro-regions, indicating substantial heterogeneity in local economic conditions.

4.3 Market-Level Results

I begin by documenting the relationship between local exposure and changes in employment. Table 1 compares micro-regions in each of the three exposure groups to those with both weak potential demand and weak enforcement capacity. Column (1) shows that the reform led to a significant 1.1 percentage point (p.p.) increase in the share of people with disabilities employed in the formal sector, but only in local labor markets with both strong potential demand and enforcement capacity. This effect corresponds to a 9.5 percent increase relative to the baseline mean of 0.117 in 2000. In other words, moving a region from the lower median (e.g., Rio Negro in the Amazon) to the upper median (e.g., São Paulo) of the enforcement–demand distribution raises formal employment among people with disabilities by roughly 9.5 percent. Conversely, employment remains unaffected in areas where either demand or enforcement is weak, underscoring the strong complementarities between regulation and enforcement.

As placebo tests, Columns (2)–(4) assess whether exposure generated negative spillovers, such as the displacement of workers without disabilities, or prompted shifts in the informal sector. The coefficients are small and statistically insignificant across all specifications. The absence of significant effects on workers without disabilities suggests a lack of crowding out, indicating that the reform induced net formal job creation rather than a substitution across

worker types. Furthermore, the lack of changes in informality rates suggests that the rise in formal employment does not stem from a simple reallocation from informal to formal jobs.

Turning to earnings, Table 2 shows that only micro-regions with both strong potential demand and enforcement capacity experience increases in work income for people with disabilities. The average work income rises by 7 percent relative to the baseline mean in 2000, a result primarily driven by the extensive margin of higher employment. There are no spillovers to non-work income (Column (2)) or to workers without disabilities (Columns (3) and (4)). Taken together, these results point that the benefits of the reform are concentrated among people with disabilities in highly exposed areas, without creating negative spillovers for other groups.¹⁷

Additional Results. I provide two additional pieces of evidence to further validate the identification strategy. First, to address the absence of high-frequency pre-reform disability records in the Census, I conduct a pre-trend analysis using administrative data from RAIS. Although RAIS only began recording disability status in 2003, I retrospectively assign this status to individuals by identifying any worker ever reported as having a disability. Figure B5, Appendix B, corroborates the lack of differential pre-trends and validates the previous findings that people with disabilities in more exposed local labor markets experienced larger employment gains after the reform. Second, I investigate whether the observed effects were driven by spatial reallocation or intensive margin adjustments. Results in Table C3, Appendix C, show no significant changes in migration patterns or hours worked, confirming that the reform induced a net expansion of formal employment.

5 Firm-Level Analysis

The aggregate analysis shows that the disability reform increased formal employment and earnings for people with disabilities in more exposed local labor markets, without generating spillovers to workers without disabilities or to those in the informal sector. A natural next

¹⁷Table C2, Appendix C, separately considers income from formal and informal work. These definitions differ from overall work income in Table 2. Conditional on employment, I find no differences in either formal or informal earnings, confirming that the baseline gains are driven by higher formal employment rather than higher wages among the employed. This is consistent with Manning (2011), who argues that firms may respond to mandates by expanding recruitment or adjusting the skill mix of new hires rather than raising equilibrium wages. Section 5 further explores these firm-level adjustments.

question is how firms adjust internally to meet the mandated disability employment levels, since the incidence and efficiency implications of the policy depend on the specific margins through which firms adjust. To document the adjustment margins, I exploit the regulatory threshold requiring firms with at least 100 employees to employ a minimum share of workers with disabilities, together with the precise timing of labor inspections. This section presents graphical evidence of the short- and medium-term effects and examines a broad range of firm- and worker-level outcomes and mechanisms.

5.1 The Employment Effects

Empirical Strategy. In the first part of the firm-level analysis, I estimate the employment effects of disability hiring quotas. Figure D1, Appendix D, shows that the AA quota is rarely enforced in the absence of labor inspections. Because only firms with at least 100 employees are subject to the quota requirement, my empirical strategy exploits both the regulatory threshold and the precise timing of inspections. Specifically, I compare inspected firms with pre-inspection employment just above and below the 100-employee cutoff—which serve as treatment and control groups, respectively—before and after inspection. I estimate the following event-study difference-in-differences specification:¹⁸

$$y_{jt} = \sum_{k=-6}^{k=12} [\beta_k^{Quota} \times D_{jk} \times Quota_{j,pre} + \theta_k \times D_{jk}] + \alpha_j + \alpha_t + X_{jt}\gamma + \varepsilon_{jt}, \quad (4)$$

in which subscripts j and t denote firm and quarter-year; $D_{jk} = \mathbf{1}(t_j = t + k)$ are dummies indicating k quarters relative to the inspection quarter t , defined as the quarter before the final inspection report is submitted; $Quota_{j,pre} = \mathbf{1}(\text{Emp}_j \geq 100)$ is an indicator for firms with at least 100 employees prior to inspection, which identifies treated firms; α_j and α_t are firm and quarter-year fixed effects; X_{jt} includes time-varying firm-level controls and state- and industry-specific trends; and y_{jt} is the employment outcome of interest. Standard errors

¹⁸In addition to the event-study analyses, I perform aggregate difference-in-differences analyses in which I pool pre- and post-inspection periods and estimate the following model:

$$y_{jt} = \alpha_j + \alpha_t + \tilde{\beta} \times Post_t \times Quota_{j,pre} + X_{jt}\gamma + \varepsilon_{jt}, \quad (3)$$

in which subscripts and the set of controls and fixed effects are the same as in Equation (4), and $Post_t \times Quota_{j,pre}$ is an indicator variable equal to 1 for all quarters after inspection in firms under the AA quota. Standard errors are clustered at the firm level.

are clustered at the firm level.¹⁹

The post-event coefficients of interest, β_k^{Quota} , capture the dynamic effects of enforcing the quota requirement relative to two quarters before inspection. To mitigate concerns about selection into inspections, the estimation sample includes all inspected firms and focuses on each firm’s first recorded inspection. Labor inspections assess compliance with a broad set of regulations, such as formal registration, minimum wage, and mandated benefits, but only firms with at least 100 employees may also be evaluated for compliance with the AA quota. This statutory cutoff provides the key source of identifying variation.

The empirical design compares first-time inspected firms with pre-inspection employment just above and just below the 100-employee threshold. Because inspections are not targeted toward AA quota compliance, the timing of a firm’s first inspection is plausibly unrelated to its quota compliance. Any selection into inspection based on general compliance concerns should therefore affect firms on both sides of the cutoff. The differential response across the threshold can thus be attributed to the enforcement of the AA quota, rather than to inspection effects more broadly. Importantly, no other labor regulation in Brazil applied a 100-employee cutoff across all firms during the study period.²⁰ Consistent with this interpretation, Figure E1, Appendix E, shows a smooth distribution of firm size around the cutoff, suggesting that firms do not strategically sort to avoid AA quotas.

Identification in Equation (4) relies on the timing of labor inspections being uncorrelated with the outcomes of interest, conditional on firm and time fixed effects and firm-level controls X_{jt} . I provide supporting evidence showing that the timing of inspections is not systematically related to observable firm characteristics. The key identifying assumption is that, in the absence of inspection, outcomes for treated and control firms would have followed parallel trends around the inspection date. I assess this assumption by testing whether the pre-event coefficients are statistically indistinguishable from zero.

The estimates could be biased if firms in different size categories would have evolved differently without inspections. For example, macroeconomic shocks may have affected large

¹⁹A local regression discontinuity design is a natural alternative, but the running variable is poorly defined in the administrative data and some firm selection around the 100-employee cutoff may occur. Including firm and time fixed effects, along with industry and location trends, mitigates selection concerns and focuses identification on within-firm variation over time. A bunching estimator is also unsuitable. As shown in Figure D2, Appendix D, there is no visual evidence of bunching below the 100-employee cutoff. These considerations motivate the use of a dynamic difference-in-differences design as the main empirical approach.

²⁰A pay-transparency regulation (Law 14.611/2023) also applies to firms with 100 or more employees, but it was enacted after the study period.

and small firms differently. I take several steps to address this. First, the baseline specification restricts the sample to firms with pre-inspection employment between 75 and 125 workers. Second, I show robustness to narrower bandwidths and to excluding firms located very close to the 100-employee cutoff. Third, I add state- and industry-specific time-varying controls X_{jt} to absorb differential policies or unobserved shocks across states and industries.

Sample and Summary Statistics. I take several steps to construct the sample. First, I compile the universe of general inspections and retain each firm’s earliest recorded inspection. Restricting the sample to first inspections helps ensure that inspection timing is plausibly unanticipated from the firm’s perspective, as firms have no prior inspection history to predict inspection timing. Focusing on general inspections, rather than inspections explicitly targeting non-compliance, further mitigates concerns that inspections are triggered by firms’ non-compliance with disability quotas.

Second, I generate quarterly labor market information from the RAIS data by combining hiring and separation records and matching inspected firms to this quarterly panel. Third, I restrict the sample to firms observed in RAIS for six quarters before and fourteen quarters after their first inspection, allowing me to estimate dynamic effects over a five-year window around the inspection event. Fourth, I classify firms with fewer than 100 employees as controls and those with at least 100 employees as treated, based on pre-inspection employment, the same criterion used by labor inspectors to determine quota applicability.

Table F1, Appendix F, presents summary statistics. Prior to inspections, control and treatment firms are comparable along observable dimensions, including the share of workers with disabilities, average earnings, location, and economic sector. As expected, the only systematic difference is firm size: Control firms employ fewer employees on average (74.9) than treated firms (96.9). This similarity across most dimensions suggests that the timing of first inspections is not systematically related to firm characteristics and is plausibly unexpected.

Two additional pieces of evidence further support this interpretation. First, Table F2, Appendix F, shows that inspection dates and the types of violations uncovered during inspections are comparable across control and treatment firms. Second, I regress indicators for the first year of inspection on pre-inspection firm characteristics, including firm size, payroll, sector, and location, and find no systematic relationship. Taken together, these results sup-

port the identifying assumption that the timing of inspections is plausibly exogenous with respect to the firm outcomes of interest.

After inspections, employment of people with disabilities increases more sharply among treated firms. The average number of employees with disabilities rises from 0.29 to 0.77, and the share of firms employing at least one worker with disabilities increases from 13 percent before inspections to 26 percent afterward. Control firms experience only modest increases, consistent with the fact that compliance with the AA quota is not required for them.

Results: Employment of People with Disabilities. To capture the effects of enforcing the AA quota on the employment of people with disabilities, I use three complementary outcomes that capture both the extensive and intensive margins of firms’ responses: (i) an indicator variable for having at least one worker with disabilities, (ii) the number of workers with disabilities, and (iii) the share of workers with disabilities, defined as the total number of workers with disabilities divided by the total firm employment (multiplied by 100).

Across all three measures, I find strong and persistent increases in disability employment following inspections. Figures 3(a)–3(c) plot the estimated coefficients $\hat{\beta}_k^{Quota}$ and their 95 percent confidence intervals. The pre-event coefficients are statistically indistinguishable from zero, supporting the identifying assumption that treated and control firms followed similar trends prior to inspection. After inspections, disability employment rises sharply and continues to grow over time, indicating a sustained adjustment in firms’ hiring behavior.

In terms of magnitude of the intensive margin, Figure 3(a) and Column (1) of Table 3 show that inspections lead to roughly a twofold increase in the number of workers with disabilities. On the extensive margin, Figure 3(b) and Column (2) of Table 3 indicate that treated firms are 7.8 percentage points more likely to employ at least one worker with a disability. Figure 3(c) and Column (3) of Table 3 further corroborate these results, showing a significant rise in the share of workers with disabilities within firms.

I next use the panel structure of the data to examine whether these effects are driven by relabeling of existing employees with disabilities rather than genuine hiring responses. Column (2) of Table F4, Appendix F, indicates that roughly half of the 100 percent increase in the number of workers with disabilities arises from relabeling, while the remainder reflects new hires. Importantly, Figure E3, Appendix E, shows that hiring rates rise sharply and persist after inspections, demonstrating that firms adjust their recruitment behavior rather

than merely reclassifying existing workers.

Results: Other Firm Outcomes. I next examine other margins of firm adjustment to assess whether new hires with disabilities are financed through lower profits. Because direct profit measures are unavailable, I use average wages and firm exit from the formal sector as indirect proxies. Figures 3(d)–3(e) and Columns (4)–(5) of Table 3 show no significant effects on these outcomes. Figure D.2, Appendix D, reveals no bunching below the 100-employee threshold, suggesting that firms do not manipulate employment to avoid the regulation.

Turning to outcomes for workers without disabilities, Column (6) of Table 3 shows no significant decline in their employment. Although log employment displays mild pre-trends prior to inspections, these reflect mechanical differences in firm size rather than anticipatory behavior. Consistent with this interpretation, both hiring and permanent separation rates for workers without disabilities are flat in the pre-inspection period, indicating no behavioral response before treatment (Figure E3, Appendix E). Following inspections, both margins remain unchanged. Moreover, the aggregate analysis in Section 4 similarly finds no effects on workers without disabilities in more exposed local labor markets. Taken together, these results indicate that firms expand employment of workers with disabilities without displacing those without disabilities, either within firms or across local labor markets.

These findings are also consistent with related evidence showing limited efficiency costs of disability hiring quotas. For example, Mori and Sakamoto (2018) find no impact on firm profits in Japan despite increased employment of workers with disabilities under the 1.8 percent quota, and Duryea and Martínez (2023) show that the introduction of a 1 percent quota in Chile did not displace workers without disabilities.

Results: Heterogeneity. The richness of the RAIS data also allows me to examine how firms comply with the AA quota across different types of jobs. Figure E4, Appendix E, shows that the increase in disability employment is concentrated in low-skill, low-paying occupations.²¹ Firms also recruit disproportionately from individuals without a college degree.

²¹Occupational categories include: (i) managers (e.g., managers and directors); (ii) high-skill professionals (e.g., researchers, teachers, doctors, nurses, engineers, technicians, architects, mathematicians, and statisticians); (iii) low-skill white-collar jobs (e.g., cashiers, receptionists, secretaries, and library assistants); and (iv) blue-collar jobs. Consistent with these results, Holzer and Neumark (1999) find that U.S. firms subject to affirmative action are more likely to hire minorities with lower education levels and for jobs with lower skill requirements.

Another important source of heterogeneity concerns the type of disability. Since 2006, employers have been required to report whether a worker has a physical, hearing, visual, intellectual, or multiple disability (two or more disabilities). Individuals who have undergone vocational rehabilitation may also qualify as having disabilities for quota purposes. Figure E4, Appendix E, shows that employment gains are concentrated among those with physical disabilities, followed by hearing and other disabilities, indicating that firms tend to hire workers with milder forms of disability.²²

Robustness. Table F4, Appendix F, presents additional checks showing that the results are robust to alternative variables, model specifications, and sample definitions. Column (1) reproduces the benchmark estimates from Table 3 for comparison. Columns (3)–(4) show that the main conclusions regarding the employment of people with disabilities are unchanged when using the inverse hyperbolic sine transformation or a conditional fixed-effects Poisson model to account for the count nature of the dependent variable. The Poisson estimates also confirm the absence of significant negative effects for workers without disabilities (Column (5)). Column (6) reports similar results when excluding state- and industry-specific trends. Finally, Columns (7)–(10) examine the sensitivity of the results to the sample definition: estimates remain stable when restricting the sample to narrower bandwidths around the 100-employee threshold or when excluding firms located close to it.

5.2 The Worker Effects

Empirical Strategy. While the finding that AA quotas led to higher employment for the targeted group is consistent with the literature, there is relatively little evidence of their impacts on workers’ outcomes, including wages. To further understand these impacts, I estimate the following event-study difference-in-differences model:²³

²²Because visual, intellectual, and multiple disabilities represent a small share of the total, I group them together. The RAIS data contain no additional information on disability type. For context, using administrative records from the federal public sector, the distribution of disabilities among employed individuals is as follows: partial visual impairment (16.7%), congenital or acquired deformity (13.1%), reduced mobility (10.2%), partial hearing impairment (8.5%), bilateral hearing impairment (6.0%), monoparesis (5.3%), amputation (5.0%), deafness (4.7%), paraplegia (3.7%), monoplegia (3.1%), blindness (2.8%), hemiparesis (1.8%), paraparesis (1.7%), multiple disabilities (1.2%), tetraparesis (1.0%), cerebral palsy (1.0%), hemiplegia (0.9%), intellectual disability (0.7%), tetraplegia (0.5%), dwarfism (0.4%), and others.

²³In addition to the event-study analysis, I conduct a difference-in-differences analysis in which I pool pre- and post-inspection quarters and estimate the change in wages considering the following model:

$$y_{ijt} = \sum_{k=-6}^{k=16} [\beta_k^{Quota} \times D_{jk} \times Quota_{j,-1} + \theta_k \times D_{jk}] + \alpha_j + \alpha_t + X_{jt}\gamma + X_{ijt}\delta + \varepsilon_{ijt}, \quad (6)$$

in which subscripts i , j , and t denote worker, firm, and quarter-year, respectively. $D_{jk} = \mathbf{1}(t_j = t + k)$ are dummies indicating an event in quarter k relative to the quarter t of the firm's inspection. $Quota_{j,-1} = \mathbf{1}(\text{Emp}j \geq 100)$ indicates firms subject to the AA quota in the quarter before inspection. α_j and α_t denote firm and quarter fixed effects, and y_{ijt} is the worker outcome of interest. The vectors X_{jt} and X_{ijt} include firm- and worker-level controls. Firm-level controls consist of state- and industry-specific trends, while worker-level controls include gender, race, age, age squared, and fixed effects for educational attainment and occupation group. Standard errors are clustered at the firm level.

As in Equation (4), the post-event coefficients β_k^{Quota} capture the dynamic effects of the AA quota relative to the quarter preceding inspection. Identification relies on the timing of labor inspections being uncorrelated with workers' outcomes, conditional on the included controls. The key identifying assumption is that, absent inspection, workers' outcomes in treated and control firms would have evolved in parallel for $k > 0$. I assess this assumption by testing whether the pre-event coefficients are statistically indistinguishable from zero.

Sample and Summary Statistics. To examine the wage effects, I use the natural logarithm of hourly wages as the main outcome.²⁴ The sample consists of all workers employed by the same set of firms used in the firm-level analysis. I construct the worker-level panel at a quarterly frequency, spanning six quarters before and sixteen quarters after inspection.²⁵

This structure offers two advantages. First, it ensures comparability with the firm-level analysis, allowing us to assess how firm adjustments translate into worker outcomes. Second, the longer panel increases the sample size and statistical power, a relevant feature given the

$$y_{ijt} = \tilde{\beta} \times Post_t \times Quota_{j,-1} + \alpha_j + \alpha_t + X_{jt}\gamma + X_{ijt}\delta + \varepsilon_{ijt}, \quad (5)$$

in which subscripts and the set of controls and fixed effects are the same as in Equation (6), and $Post_t \times Quota_{j,-1}$ is an indicator variable equal to 1 for all quarters after inspection in firms under AA quota. As before, standard errors are clustered at the firm level.

²⁴This variable combines information on contracted hours and average wages, which include other forms of compensation such as overtime premiums, bonuses, commissions, and benefits.

²⁵Until 2015, the RAIS data report wages annually. I convert them to quarterly observations by combining information on total earnings with hiring and separation dates.

relatively low employment rates among people with disabilities.

Table F5, Appendix F, presents summary statistics for treated and control firms before and after inspection. Workers in both groups are similar along observable characteristics, including disability status, wages, gender, race, education, occupation, location, and sector of activity. In the pre-inspection period, about 0.4 (0.3) percent of workers in control (treated) firms have a disability; average hourly earnings are 12.72 (12.42) Brazilian *reais*; 66 (65) percent are male; 66 (68) percent are white; 10 (10) percent hold a college degree; and 66 (66) percent work in blue-collar occupations.

Results: The Wage Effects. Figure 4(a) plots the wage effects of enforcing the AA quota on wages for workers with and without disabilities after estimating Equation (6), which compares employees at firms subject to the quota with those at control firms before and after inspection. The pre-event coefficients are statistically indistinguishable from zero, supporting the identifying assumption of parallel trends. As expected, confidence intervals for workers with disabilities are relatively wider because of the smaller sample size.

Table 4 reports the immediate ($k = 0$), short-run ($k = 6$), and long-run ($k = 12$) effects, as well as the average post-inspection impact. In the quarters immediately following inspection, wage trajectories diverge for both groups. While wage growth for workers without disabilities remains flat, workers with disabilities experience a persistent slowdown. The average post-inspection estimate implies a 5.6 percent decline in wage growth for workers with disabilities (Column (4)). Because the specification includes firm and time fixed effects, identification comes from within-firm comparisons over time and across disability status.

Columns (5)–(6) distinguish between incumbent and newly hired workers. Wage growth declines for both groups, although the estimates for new hires are less precise. Taken together with the employment results, this pattern is consistent with firms partially offsetting the cost of compliance by slowing wage growth for incumbent and new workers with disabilities, thereby widening within-firm wage inequality between workers with and without disabilities.

The observed slowdown in wage growth could mechanically reflect differences in education, experience, or occupational sorting across groups. However, controlling for these characteristics leaves the estimated effects essentially unchanged (Column (2)). Moreover, reweighting workers with and without disabilities using inverse propensity score and entropy-balancing methods (Hainmueller, 2012) yields similar results (Figure E5, Appendix E). These

findings reassures that the wage effects are not driven by compositional differences.

Results: Other Worker Outcomes. Figure 4 and Table 5 show that quota enforcement affects the retention margin for the targeted group but leaves the intensive margin of employment unchanged. Figures 4(b) and 4(c) point to no significant effects on the intensive margin, measured by contractual hours and the probability of part-time arrangements, for workers with and without disabilities. Figure 4(d) reports that workers with disabilities at firms under the quota are more likely to remain employed with the same firm in the quarters following inspections. This increase in job retention is temporary and gradually dissipates over time. No comparable effect appears for workers without disabilities.

What Can Explain the Wage Effects? Why do workers with disabilities experience lower wage growth? I argue that the results are consistent with two complementary labor market frictions: imperfect competition and discrimination.

Workers with disabilities face more limited outside options and greater mobility and search frictions, reducing the elasticity of their labor supply to the firm. In such environments, employers have wage-setting power and can comply with disability quotas by expanding employment while keeping wage growth for workers with disabilities relatively low. Consistent with reduced outside options, my previous results indicate that workers with disabilities exhibit higher retention rates despite experiencing lower wage growth.

To provide more direct evidence of imperfect competition as a relevant friction, I examine whether the slowdown for workers with disabilities is stronger in more concentrated local labor markets, where search frictions are more severe and firms have greater monopsony power. I construct a Herfindahl–Hirschman Index (HHI) of employer concentration across microregions and estimate heterogeneity of Equation (6) by splitting markets at the median of the HHI distribution. If imperfect competition is relevant, wage effects should be more pronounced in above-median HHI markets, where competitive pressure on wages is weaker. Columns (1) and (2) of Table 6, Panels A and B, support this prediction. Reassuringly, Panels C and D indicate no heterogeneity for workers without disabilities, suggesting that these patterns do not reflect broader regional trends.

The results are also consistent with dynamic statistical discrimination. A large literature documents discrimination against people with disabilities using observational and experimen-

tal evidence (Baldwin and Johnson, 1994, 2006; Baert, 2016; Ameri et al., 2018). Models of statistical discrimination posit that employers observe imperfect, noisy, imperfect signals of worker productivity and partially rely on group averages when forming expectations (Phelps, 1972; Arrow, 1973; Aigner and Cain, 1977). In dynamic settings, if signals from a minority group are less informative, members of that group must accumulate more positive evidence to access higher-level tasks and promotions, producing slower wage growth (Bjerk, 2008). Affirmative action can reinforce this mechanism by increasing hiring from the minority group and lowering the average informativeness of their signals (Lehmann, 2011).²⁶

In my setting, AA quotas may induce firms to lower hiring thresholds for people with disabilities, consistent with models where affirmative action reduces the precision of market signals (Coate and Loury, 1993; Moro and Norman, 2003; Fang and Moro, 2011). Administrative and survey evidence support this: firms under AA quotas recruit more disabled workers without a college degree (Figure E4, Appendix E), and the HR survey respondents report greater difficulty finding qualified applicants with disabilities (Table G2, Appendix G). A central implication of dynamic statistical discrimination is that workers whose skills are harder to signal should experience slower wage growth. I propose an indirect test, which leverages heterogeneity of Equation (6) using two observable proxies for signal from the administrative data: formal education and the degree of interpersonal interaction required in the job.²⁷ Columns (3)–(6) of Table 6 show wage growth declines under the quota are concentrated among workers with disabilities with weaker opportunities to signal ability—those without a high school diploma and those in low-contact occupations. Workers with disabilities in high-signal occupations do not experience comparable declines.

Can Other Mechanisms Explain the Wage Effects? A number of alternative mechanisms could, in principle, generate lower wage growth for workers with disabilities. While these channels may contribute, several pieces of evidence suggest they are unlikely to be the primary drivers of the results.

First, taste-based discrimination is unlikely to explain wage dynamics for workers with

²⁶As examples, Bjerk (2008) develop a “sticky-floor” model in which women and minorities progress more slowly even without taste-based discrimination. Lehmann (2011) show that affirmative action in U.S. law firms increased hiring of Black associates while limiting partnership prospects because firms required stronger evidence of ability.

²⁷Education information come from RAIS. Interpersonal contact is based on O*NET measures matched to RAIS occupations. We define occupations above the median in required contact are classified as offering stronger opportunities to signal skills.

disabilities. Taste-based models posit that employers have a negative animus towards people with disabilities, regardless of productivity (Becker, 1957). If such animus is binding, an firms subject to the quota would be expected to hire *more qualified* workers with disabilities. Instead, I find that the employment increases are concentrated among workers without a college degree. In addition, to explain the sharp decline in wage growth, taste-based discrimination would have to strengthen quite rapidly, a pattern at odds with evidence that underlying tastes are relatively stable in the short run (Beaman et al., 2009)

Second, fixed-cost arguments, such as the idea that firms incur substantial expenses to accommodate or screen workers with disabilities (Oi, 1991; Acemoglu and Angrist, 2001; Miller, 2017), are also inconsistent with the data. Incumbent workers with disabilities were employed before the inflow of new hires, suggesting that any such costs would have been borne earlier. In addition, survey evidence from Appendix G suggests that these costs are minimal: only 8 percent of firms reported providing workplace accommodations to their most recent hire with disabilities, and none report higher costs relative to hiring a comparable worker without disabilities.²⁸ The lack of evidence of costly workplace accommodations or other amenities also makes compensating differentials an unlikely explanation (Rosen, 1986).

Third, rent-sharing models suggest that the quota could reduce firm rents and therefore wage growth (Stole and Zwiebel, 1996; Jäger and Heining, 2022). However, the absence of significant effects on firm exit, profits, or the wages of workers without disabilities, as well as the robustness of the results to reweighting methods, suggests that reduced rents are unlikely to explain the wage patterns for workers with disabilities.

Finally, retaliation against suspected whistleblowers does not appear to be a plausible explanation for the wage patterns observed among employees with disabilities. Violations of labor regulations are widespread and not specific to compliance with disability quotas: 42 percent of firms have registration violations, 20.2 percent violations related to severance pay, and 17.5 and 13.0 percent violations related to working hours and days off, respectively. Such pervasive infractions make it unlikely that firms systematically associate workers with disabilities with whistleblowing. Moreover, while 41.7 percent of HR survey respondents report that inspections are often triggered by employee complaints, none identified workers with disabilities as the source of such complaints.

²⁸As a benchmark, the U.S. Job Accommodation Network survey finds that 56 percent of accommodations cost nothing, 39 percent required a one-time expense, and only 5 percent generated annual costs, with a median one-time cost of \$500 (Job Accommodation Network, 2020).

6 Implications and Discussion

Having established the market- and firm-level impacts of enforcing AA quotas, I now turn to their welfare implications for firms, workers, and the government. The empirical results show that enforcement raises employment of workers with disabilities without generating displacement or broader labor market spillovers, motivating a partial-equilibrium welfare analysis. To guide this analysis, Appendix H develops a parsimonious model that maps firms' compliance decisions under inspections into changes in employment, producer surplus, worker surplus, and fiscal revenues. This framework provides the minimal structure needed for welfare analysis using the estimated effects in the data.

Firms. On the firms' side, Section 5 shows that firms subject to AA quotas are no more likely to exit the formal sector, reduce wages for workers without disabilities, or bunch below the regulatory threshold. These patterns suggest that firm profits remain stable. To corroborate this interpretation in the absence of profit data, I develop a simple discrete-choice framework that delivers a closed-form expression for the marginal revenue product ($MRPL_d$) of newly hired workers with disabilities. This exercise also speaks to a broader gap in the literature: the scarcity of estimates of productivity for this group.

Conditional on inspection and detection, non-compliant firms can choose between paying the statutory fine or complying by hiring the required number of workers. If it chooses to pay fines, the firm i gets utility $U_{i,f} = -\gamma_i f_i$, where f_i is the statutory fine per missing worker and γ_i is the quota shortfall.²⁹ If the firm i chooses compliance, it obtains utility $U_{i,c} = \gamma_i(MRPL_d - w_i - \eta_i)$, in which $MRPL_d$ is the marginal revenue product of newly hired workers with disabilities, w_i is the wage paid to these workers, and $\eta_i \sim N(0, \sigma^2)$ captures unobserved per-worker (real and perceived) hiring costs not reflected in wages, such as workplace accommodations, job redesign, additional supervision, or taste-based discrimination. The probability of compliance is therefore:

$$P_{i,c} = \Pr(U_{i,c} \geq U_{i,f}) = \Phi(\alpha + \beta_1 w_i + \beta_2 f_i), \quad (7)$$

where $\alpha \equiv \frac{MRPL_d}{\sigma}$; $\beta_1 \equiv -\frac{1}{\sigma}$; and $\beta_2 \equiv \frac{1}{\sigma}$. A probit model yields consistent estimates under standard assumptions, and a logistic specification produces close results. Because

²⁹The quota shortfall component is the gap between the potential number of jobs reserved by the quota regulation, which is calculated from the firm size, and the actual number of employees with disabilities.

$MRPL_d = -\frac{\alpha}{\beta_1}$, the marginal revenue product of workers with disabilities hired under the quota is identified directly from the estimated coefficients.

Identification relies on the statutory nature of wages and fines, which provides variation plausibly exogenous to a firm’s unobserved hiring costs. For compliant firms, I compute the counterfactual fine they would have paid had they remained non-compliant using the statutory fine schedule from Ordinance 1,199/2003. In practice, I construct predicted fines based exclusively on predetermined enforcement components, including firm size brackets, local labor office, economic sector, and year. This approach isolates the institutional, policy-driven component of fines and purges any firm-specific variation that could be correlated with unobserved compliance costs. For firms that opt to pay fines, I impute the counterfactual wage they would have paid had they hired workers with disabilities using the observed distribution of wages of new hires without disabilities in similar low-skill white- and blue-collar jobs. This approach is supported by Brazilian labor law, which mandates equal pay for new hires performing identical jobs, and by my finding in Section 5 that workers with disabilities are primarily hired into these jobs. Because both fines and wages are determined by regulations and institutional rules rather than individual firm preferences, they are plausibly orthogonal to the unobserved cost term η_i , satisfying the exclusion restriction required for identification.

The estimates in Table 7 imply that the ratio of the marginal revenue product of disabled workers to median wages is approximately 1.30–1.55. This means that newly hired workers with disabilities generate output at least commensurate with their wage cost.³⁰

Workers. On the workers’ side, empirical results show that people without disabilities are unaffected in terms of employment or earnings, so their economic surplus remains unchanged. For newly hired workers with disabilities, employment generates worker surplus equal to wage income net of the worker’s reservation value of time, inclusive of any opportunity costs of work. In a perfectly competitive labor market, such surplus would be zero. However, the high involuntary unemployment rates among people with disabilities (Column (2) of Table A1, Appendix A) and the presence of substantial frictions in the Brazilian labor

³⁰There could be an opportunity cost if firms would otherwise have hired a worker without disabilities instead of a worker with disabilities. Such substitution would reduce surplus by the difference in their marginal products. In practice, however, Figure E3, Appendix E, shows no evidence of labor–labor substitution in the form of permanent separation or reduced hiring of workers without disabilities. This pattern indicates that large firms adjust by recruiting workers with disabilities rather than substituting across worker types, suggesting that the opportunity cost from foregone hires without disabilities is likely negligible.

market are inconsistent with a perfectly competitive environment, implying that employment relationships generate positive surplus for workers with disabilities. For incumbent workers with disabilities, the modest reduction in wage growth reflects a redistribution from workers to firms and therefore constitutes a transfer rather than an efficiency loss.

To translate these welfare gains into monetary units, I adopt a conservative, money-metric approach and impose three restrictive assumptions. First, I assume linear utility in income, so that changes in utility map directly into changes in net earnings. Second, I restrict the surplus from employment to accrue only in the first year of the spell, abstracting from any longer-run effects on earnings or attachment. Third, I adjust for the opportunity cost of work. Following Mas and Pallais (2017), I value lost leisure at 58 percent of the wage. Under these assumptions, the reduced-form estimates imply that each inspection generates a positive marginal economic surplus for people with disabilities of approximately 771 Brazilian *reais*, roughly half of the typical entry level monthly wage for workers in the affected jobs. Appendix H details the full calculation.

Government. On the government side, the relevant objects are the cost of enforcement and the fiscal savings that arise when previously non-employed individuals with disabilities enter work. While there is no data on the variable cost of each inspection, aggregate expenditures on enforcement capacity imply an average cost of inspections of around 99.29 Brazilian *reais*, which provides an upper bound for the marginal cost of enforcement. The fiscal savings arise from higher income-tax revenues and lower welfare expenditures associated with the increase in employment among people with disabilities, and amount to 856 Brazilian *reais* per inspection, roughly 8.5 times the cost of enforcement.³¹

Table H1, Appendix H, also reports the break-even ratio between the marginal revenue product of labor and average wages—that is, the minimum productivity level at which the welfare gains from enforcement justify its costs. Across all scenarios, including those that exclude the opportunity cost of work and fiscal savings, this ratio remains below one.

Discussion. If the surplus from employing workers with disabilities is positive and firms do not incur detectable losses, what friction prevents employers from voluntarily hiring these workers in the absence of regulatory pressure? The welfare analysis indicates that the barrier

³¹I note that these figures represent a lower bound of the government’s true fiscal benefits, as they exclude the reduction in distortionary taxation.

lies at the hiring margin, not in the production relationship itself.

Conceptually, the results are most consistent with discrimination at the hiring margin. A substantial body of evidence documents employer discrimination against people with disabilities in both observational and experimental settings across many contexts ([Baldwin and Johnson, 1994, 2006](#); [Baert, 2016](#); [Ameri et al., 2018](#)). Brazil is no exception. My vignette experiment shows that HR personnel are 22.2 percent less likely to express interest in an otherwise identical candidate with a disability unrelated to productivity and requiring no workplace accommodation.³² This magnitude is consistent with audit evidence from the United States, where candidates with disabilities receive substantially fewer interview callbacks than observationally equivalent candidates without disabilities ([Ameri et al., 2018](#)), and from Canada, where revealing a disability reduces callback rates by 25 percentage points and stronger quality signals do not close the gap ([Bellemare et al., 2023](#)). Taken together, this evidence establishes that the reluctance to hire workers with disabilities is not explained by productivity concerns, as it persists even when productivity is held constant or signaled clearly. Discrimination makes firms reluctant to hire workers with disabilities voluntarily, even when their marginal products are comparable to or exceed their wages. In such labor markets, hiring quotas can expand employment for the targeted group without displacing workers without disabilities or adversely affecting firm outcomes.

Search and informational frictions could also play a role. For instance, firms in my survey report difficulty finding qualified candidates with disabilities, suggesting that this difficulty could suppress hiring even absent bias. But these frictions are more likely to reinforce discrimination than to replace it as the primary explanation. Importantly, they do not alter the central finding: Firms are leaving surplus from employment relationships unrealized. These frictions also point to substantial hiring barriers that contribute to the persistent and inefficient underemployment of people with disabilities.

The results speak more broadly to a class of labor market frictions in which discrimination at the hiring margin limits employment even when matches would generate positive surplus. In such settings, modest hiring quotas can, in principle, expand employment for minorities that face similar barriers, such as people with criminal records and long-term

³²Respondents were shown a vignette describing a fictitious consulting firm seeking an entry-level clerical worker. The candidate was a 22-year-old high school graduate with strong organizational and interpersonal skills. For a random subset of respondents, the vignette additionally stated that the candidate had bilateral hearing loss. Respondents then rated, on a four-point scale, how likely the firm would be to hire him. [Table G1](#) reports the results.

unemployed workers, without imposing significant efficiency losses. This possibility is especially relevant because existing empirical evidence indicates that standard wage subsidies for these populations tend to be ineffectual at overcoming barriers at the hiring stage (Katz, 1996; Cullen et al., 2023; Jain et al., 2025). Investigating whether modest hiring mandates could be an alternative pathway to improve labor market outcomes for these disadvantaged groups remains an important question for future research.

7 Conclusion

This paper provides a novel and comprehensive assessment of the impacts of enforcing modest quotas for one of the most disadvantaged yet understudied groups: people with disabilities. Exploiting the timing of a reform in Brazil that strengthened the enforcement of a quota regulation, my market-level results indicates that people with disabilities living in local labor markets more exposed to the reform experience larger increases in formal sector employment and earnings than those in less exposed areas. Using variation in enforcement generated by labor inspections, I show that the employment gains induced by the quotas do not come at a discernible cost to other workers without disabilities in terms of wages or employment, nor do they impose detectable burdens on firms. Instead, they increase fiscal revenues. The results together indicate that, in labor markets characterized by discrimination at the hiring level, enforcing modest hiring quotas can generate efficiency gains.

This paper offers important insights into a regulation used in more than 115 countries but for which comprehensive empirical evidence has been relatively scarce. One limitation of my setting is that the firm-level analysis is based on modest quotas of 2 percent. Although this threshold lies within the typical range of disability mandates worldwide, the available variation does not allow me to study how firms and workers would respond under substantially larger requirements. The estimates primarily capture hiring at the margin among individuals with some labor force attachment and relatively mild disabilities, whereas more aggressive quotas could require firms to hire deeper into the disability distribution, where productivity, participation, and accommodation needs may differ. At the same time, the modest size of the quota helps limit concerns about general equilibrium spillovers: I find no effects on the employment or wages of workers without disabilities, no increases in informal employment, and no changes in firm survival, implying that adjustments occur largely within treated firms

rather than through broader market reallocation.

Disability hiring quotas might also generate other positive externalities beyond the direct job surplus documented in this paper. For instance, firms fostering an inclusive environment might benefit from higher diversity in teams of workers and complementarities between workers in production. In addition, increased employment among people with disabilities could generate broader social benefits, such as improvements in health, reductions in mortality, better access to health services, or lower involvement in criminal activities. Examining these additional externalities is left for future work. This paper should therefore be viewed as an initial step toward characterizing the social benefits of boosting disability employment.

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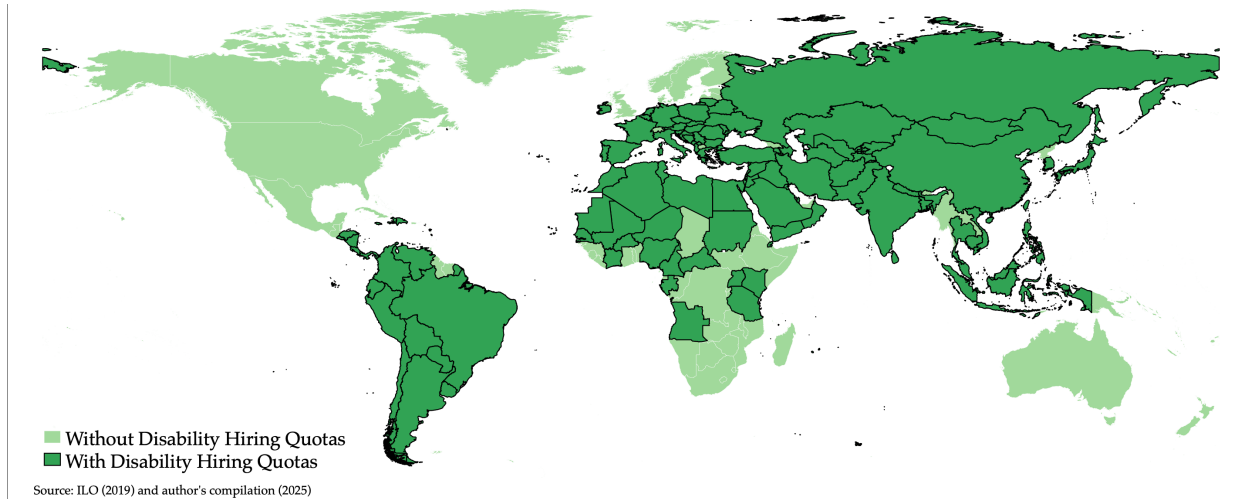
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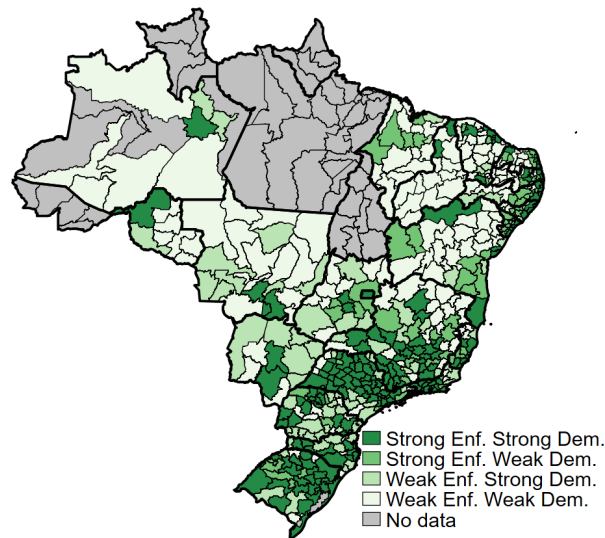
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Figure 1: Countries with Disability Hiring Quotas for People with Disabilities



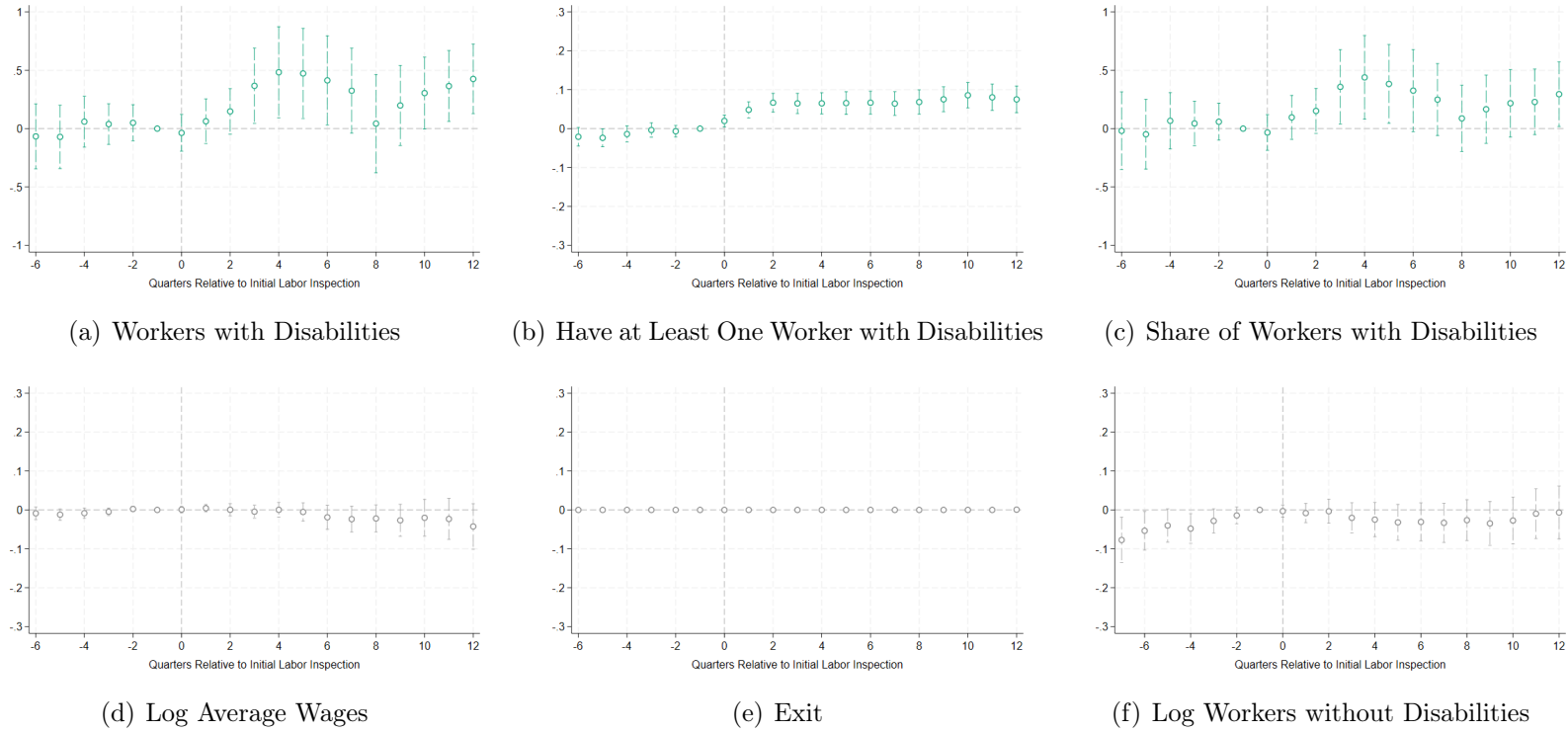
Note: This map shows the distribution of affirmative action hiring quotas for people with disabilities, implemented in more than 115 countries. Source: [ILO \(2019\)](#) and author's compilation in 2025.

Figure 2: Cross-Market Variation: Interaction between Potential Demand & Enforcement



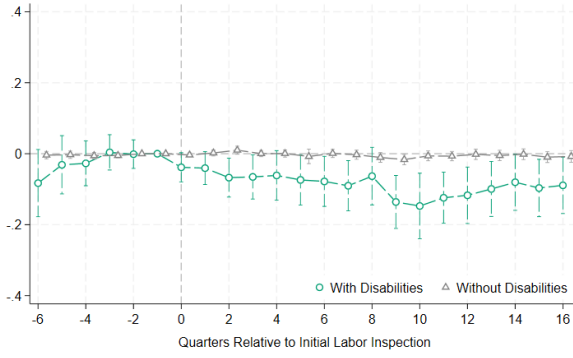
Note: This graph illustrates the geographic variation in the interaction between potential demand and enforcement capacity across micro-regions. Potential demand is defined as the total number of private-sector jobs available to people with disabilities under perfect compliance with the AA quota in 1998, calculated from the distribution of firm size and divided by the total number of people with disabilities in 2000. Enforcement capacity is proxied by the minimum distance to the nearest labor office within the same state for each micro-region. Sources: 1998 RAIS, 2000 Demographic Census, and labor office addresses.

Figure 3: The Firm-Level Effects of AA Quotas

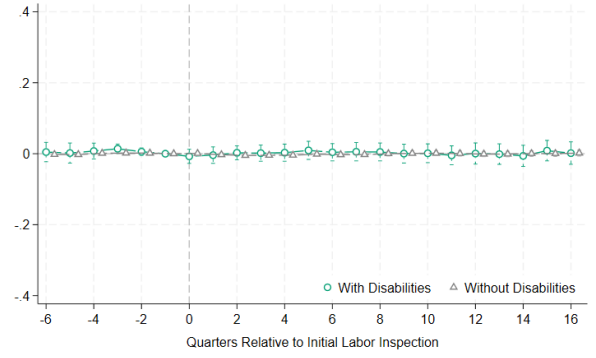


Note: This figure reports point estimates of the quarterly effects of the AA quota on the employment of people with disabilities, considering the intensive margin (total number of workers with disabilities), the extensive margin (an indicator for having at least one worker with disabilities), and the share of workers with disabilities (defined as the number of workers with disabilities divided by the total number of workers). In the bottom panel, the figure reports point estimates of the quarterly effects of the AA quota on log average wages, exit, and the log number of workers without disabilities. More details are provided in Table 3.

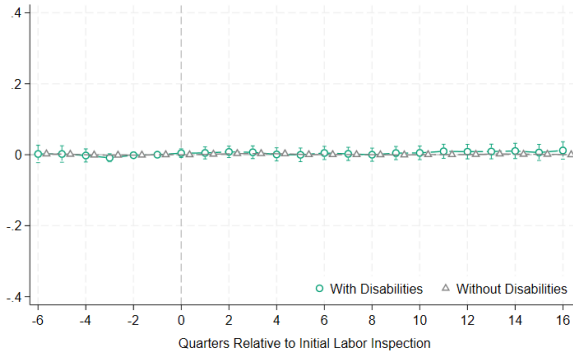
Figure 4: The Worker-Level Effects of AA Quotas



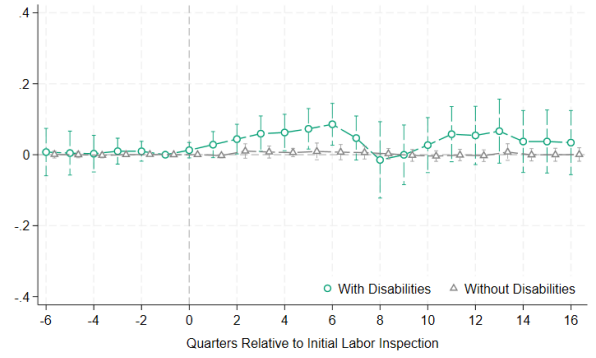
(a) Log Hourly Wages



(b) Log Number of Contracted Hours



(c) Part-Time Employment



(d) Retention

Note: This figure reports point estimates of the quarterly effects of the AA quota on log hourly wages, the log number of contracted hours, the likelihood of part-time employment, and the likelihood of remaining at the firm. More details are provided in Table 5.

Table 1: Aggregate Analysis: Impacts on Employment

	(1)	(2)	(3)	(4)
	With Disabilities		Without Disabilities	
	Formal Employment	Informal Employment	Formal Employment	Informal Employment
(Strong Demand & Strong Enforcement) $\times$ Reform	0.011*** (0.004)	-0.003 (0.004)	0.006 (0.004)	0.005 (0.004)
(Strong Demand & Weak Enforcement) $\times$ Reform	0.002 (0.004)	-0.006 (0.004)	0.002 (0.005)	0.004 (0.004)
(Weak Demand & Strong Enforcement) $\times$ Reform	0.004 (0.004)	-0.003 (0.004)	0.004 (0.004)	0.000 (0.004)
Sample Size	1,116	1,116	1,116	1,116
Mean Dep. Var. (in 2000)	0.117	0.148	0.192	0.205
Mean Dep. Var.	0.145	0.144	0.231	0.183

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the aggregate effects of the reform on formal and informal employment rates for people with disabilities (Columns (1) and (2)) and for people without disabilities (Columns (3) and (4)). All columns correspond to Equation (2). Means of the dependent variables are computed across all micro-regions. Standard errors are clustered at the micro-region level.

Table 2: Aggregate Analysis: Impacts on Earnings

	(1)	(2)	(3)	(4)
	With Disabilities		Without Disabilities	
	Work Income	Non-Work Income	Work Income	Non-Work Income
(Strong Demand & Strong Enforcement) $\times$ Reform	26.115** (12.365)	-6.033 (5.237)	9.123 (9.932)	-4.429 (3.304)
(Strong Demand & Weak Enforcement) $\times$ Reform	13.366 (20.914)	-2.051 (5.349)	7.446 (12.153)	-5.467 (4.799)
(Weak Demand & Strong Enforcement) $\times$ Reform	-10.369 (10.775)	3.836 (3.915)	6.272 (9.387)	-4.542 (3.062)
Sample Size	1,116	1,116	1,116	1,116
Mean Dep. Var. (in 2000)	367.17	119.88	638.17	70.23
Mean Dep. Var.	401.98	142.94	689.70	81.33

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the aggregate effects of the reform on earnings, measured as work and non-work income, for people with disabilities (Columns (1) and (2)) and for people without disabilities (Columns (3) and (4)). All columns correspond to Equation (2). Means of dependent variables are computed across all micro-regions. Standard errors are clustered at the micro-region level.

Table 3: Effects of AA Quotas on Employment Outcomes

	(1) Workers with Disabilities	(2) Have at Least One Worker with Disabilities	(3) Share of Workers with Disabilities	(4) Log Average Wages	(5) Exit	(6) Log Workers without Disabilities
Panel A: Dynamic Impacts						
Immediate ($k = 0$)	-0.037 (0.080)	0.020** (0.008)	-0.033 (0.078)	0.001 (0.004)	-0.000 (0.000)	-0.003 (0.008)
Short Run ($k = 6$)	0.413** (0.195)	0.067*** (0.015)	0.326* (0.179)	-0.019 (0.016)	0.000 (0.000)	-0.031 (0.025)
Long Run ($k = 12$)	0.426*** (0.152)	0.075*** (0.017)	0.295** (0.141)	-0.042 (0.030)	0.001 (0.001)	-0.007 (0.035)
Panel B: Aggregate Impacts						
Post $\times$ Quota	0.308*** (0.109)	0.078*** (0.011)	0.231** (0.104)	-0.010 (0.012)	0.000 (0.000)	0.010 (0.024)
Sample Size	65,440	65,440	65,440	65,440	76,940	65,440
Firm and Year FEs	✓	✓	✓	✓	✓	✓
State and Industry Trends	✓	✓	✓	✓	✓	✓
# Firms	3,279	3,279	3,279	3,279	3,856	3,279
Mean Dep. Var (Control)	0.310	0.087	0.383	7.474	0	4.235

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the firm-level effects of the AA quota on several outcomes: the total number of workers with disabilities, an indicator for having at least one worker with disabilities, the share of workers with disabilities relative to the total number of workers, log average wages, firm exit, and the log number of workers without disabilities. All columns correspond to Equation (4). Means of the dependent variables are computed from the control group in the quarterly window $[-6, -1]$ before the first inspection. Standard errors are clustered at the firm level.

Table 4: Effects of AA Quotas on Wages

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dynamic Impacts (With Disabilities) (N = 41,218)						
Immediate ($k = 0$)	-0.028 (0.023)	-0.033 (0.022)	-0.031 (0.022)	-0.033 (0.022)	-0.035 (0.016)	0.140 (0.143)
Short Run ($k = 6$)	-0.102** (0.042)	-0.081** (0.037)	-0.100** (0.042)	-0.081** (0.037)	-0.060 (0.037)	-0.053 (0.127)
Long Run ($k = 12$)	-0.133*** (0.045)	-0.121*** (0.043)	-0.135*** (0.044)	-0.123*** (0.041)	-0.087* (0.050)	-0.019 (0.133)
Panel B: Aggregate Impacts (With Disabilities) (N = 41,218)						
Post $\times$ Quota	-0.072** (0.033)	-0.060* (0.031)	-0.068** (0.033)	-0.058* (0.031)	-0.046 (0.034)	-0.068 (0.076)
Panel C: Dynamic Impacts (Without Disabilities) (N = 7,937,919)						
Immediate ($k = 0$)	-0.002 (0.003)	-0.003 (0.003)	-0.002 (0.003)	-0.003 (0.003)	-0.001 (0.003)	0.003 (0.012)
Short Run ($k = 6$)	0.005 (0.006)	0.005 (0.006)	0.001 (0.006)	0.002 (0.006)	0.003 (0.007)	-0.003 (0.014)
Long Run ($k = 12$)	-0.002 (0.008)	-0.002 (0.008)	-0.001 (0.008)	-0.002 (0.007)	0.005 (0.008)	-0.004 (0.020)
Panel D: Aggregate Impacts (Without Disabilities) (N = 7,937,919)						
Post $\times$ Quota	0.000 (0.005)	0.002 (0.005)	-0.002 (0.005)	-0.000 (0.005)	0.000 (0.005)	-0.002 (0.008)
Firm and Quarter FEs	✓	✓	✓	✓	✓	✓
State and Industry Trends	✓	✓	✓	✓	✓	✓
Individual Controls		✓		✓		
Occupation FE			✓	✓		
Worker FE					✓	
Sample						New Hires

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the worker-level effects of the AA quota on log hourly wages for workers with disabilities (Panels A and B) and without disabilities (Panels C and D). Panels A and C present event-study difference-in-differences estimates from Equation (5), while Panels B and D report aggregate difference-in-differences estimates from Equation (6). Column (1) includes firm and quarter fixed effects as well as state- and industry-specific trends. Column (2) adds individual controls (gender, race, educational level fixed effects, age, and age squared). Column (3) includes occupation fixed effects. Column (4) presents the preferred specification described in Equation (6). Column (5) includes worker, firm, and quarter fixed effects, along with state and industry trends. Column (6) uses the same specification as Column (4) but restricts the sample to new hires. Standard errors are clustered at the firm level.

Table 5: Effects of AA Quotas on Workers' Outcomes

	(1) Log Hours	(2) Part-Time	(3) Retention
Panel A: Dynamic Impacts (With Disabilities) (N = 41,218)			
Immediate ($k = 0$)	-0.005 (0.011)	0.002 (0.006)	0.011 (0.011)
Short Run ($k = 6$)	0.005 (0.013)	0.003 (0.009)	0.077*** (0.029)
Long Run ($k = 12$)	0.003 (0.016)	0.005 (0.010)	0.050 (0.040)
Panel B: Aggregate Impacts (With Disabilities) (N = 41,218)			
Post $\times$ Quota	0.002 (0.013)	0.002 (0.009)	0.032 (0.024)
Panel C: Dynamic Impacts (Without Disabilities) (N = 7,937,919)			
Immediate ($k = 0$)	0.001 (0.001)	-0.000 (0.001)	-0.001 (0.002)
Short Run ($k = 6$)	-0.003 (0.003)	0.001 (0.002)	0.001 (0.008)
Long Run ($k = 12$)	-0.001 (0.004)	0.001 (0.002)	-0.005 (0.008)
Panel D: Aggregate Impacts (Without Disabilities) (N = 7,937,919)			
Post $\times$ Quota	-0.001 (0.003)	0.001 (0.002)	0.000 (0.005)
Firm and Quarter FEs	✓	✓	✓
State and Industry Trends	✓	✓	✓
Individual Controls	✓	✓	
Occupation FE	✓	✓	
Worker FE			✓

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the worker-level effects of the AA quota on log contracted hours, the likelihood of having a part-time job, and the likelihood of retention. Results are presented separately for workers with disabilities (Panels A and B) and without disabilities (Panels C and D). Panels A and C use the preferred specification described in Equation (6), whereas Panels B and D correspond to Equation (5). The only exception is Column (3): the specification for the likelihood of staying at the firm includes worker, firm, and quarter fixed effects, along with state- and industry-specific trends. Standard errors are clustered at the firm level.

Table 6: Heterogeneity: Effects of AA Quotas on Wages

	(1)	(2)	(3)	(4)	(5)	(6)
	HHI		Educational Level		Interpersonal Relationship	
Panel A: Dynamic Impacts (With Disabilities) (N = 41,218)						
Immediate ($k = 0$)	0.001 (0.023)	-0.039 (0.058)	-0.037 (0.023)	-0.023 (0.036)	-0.048** (0.022)	-0.040 (0.033)
Short Run ($k = 6$)	-0.069 (0.054)	-0.093 (0.066)	-0.107* (0.055)	-0.036 (0.055)	-0.102 (0.063)	-0.054 (0.047)
Long Run ($k = 12$)	-0.074 (0.058)	-0.174** (0.072)	-0.145** (0.065)	-0.034 (0.059)	-0.151** (0.063)	-0.063 (0.055)
Panel B: Aggregate Impacts (With Disabilities) (N = 41,218)						
Post $\times$ Quota	-0.009 (0.044)	-0.127** (0.060)	-0.083* (0.045)	0.026 (0.046)	-0.067 (0.047)	-0.030 (0.039)
Panel C: Dynamic Impacts (Without Disabilities) (N = 7,937,919)						
Immediate ($k = 0$)	0.001 (0.004)	-0.002 (0.005)	-0.003 (0.004)	-0.000 (0.003)	-0.000 (0.004)	-0.002 (0.004)
Short Run ($k = 6$)	-0.006 (0.008)	0.012 (0.011)	-0.005 (0.007)	0.004 (0.007)	-0.001 (0.007)	0.003 (0.008)
Long Run ($k = 12$)	-0.009 (0.010)	0.008 (0.013)	-0.003 (0.008)	-0.005 (0.008)	0.004 (0.008)	-0.007 (0.011)
Panel D: Aggregate Impacts (Without Disabilities) (N = 7,937,919)						
Post $\times$ Quota	-0.006 (0.007)	-0.001 (0.009)	-0.002 (0.005)	-0.004 (0.006)	0.002 (0.005)	-0.007 (0.007)
Sample Restriction	Low-HHI Markets	High-HHI Markets	No High School	Has High School	Contact: Low	Contact: High

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the heterogeneous effects of the AA quota on log hourly wages for workers with disabilities (Panels A and B) and without disabilities (Panels C and D). Columns (1) and (2) refer to workers in micro-regions with a Herfindahl-Hirschman Index (HHI) below and above the median of the distribution. Columns (3) and (4) restrict the samples to workers without and with a high school diploma. Columns (5) and (6) refer to samples of workers in jobs characterized by weak and strong interpersonal relationships. All columns correspond to specifications that include firm and quarter fixed effects, state- and industry-specific trends, and occupation fixed effects. Standard errors are clustered at the firm level.

Table 7: Discrete Choice Estimates

	(1)	(2)
$\hat{\beta}_0$	0.4983*** (0.0861)	1.1167*** (0.1845)
$\hat{\beta}_1$	-0.0002*** (0.0000)	-0.0005*** (0.0001)
$MR\hat{P}L_d$	2,425.002 (364.13)	2,035.465 (253.04)
Median Wages	1,555.48	1,555.48
$MR\hat{P}L_d/w$	1.5590	1.3086
[Lower Bound; Upper Bound]	[1.1002; 2.0178]	[0.9897; 1.6274]
Model	Probit	Logit

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports estimates from Equation (7) using probit (Column (1)) and logit (Column (2)) models. $MR\hat{P}L_d/\bar{w}$ is defined as the ratio between firms' estimated monthly marginal revenue product of labor and median wages. Lower and upper bounds with 95-percent confidence intervals are reported. Standard errors are computed using the delta method.

ONLINE APPENDIX

A Institutional Context and Data

A.1 The Disability Gaps in Brazil

Table A1: Disability Gaps in the Brazilian Labor Market

	(1) Economically Active	(2) Unemployment	(3) Informal Employment	(4) (IHS) Work Income
Severe Difficulty	-0.190*** (0.008)	0.064*** (0.004)	0.033*** (0.002)	-0.201*** (0.009)
Some Difficulty	-0.049*** (0.005)	0.046*** (0.002)	0.021*** (0.002)	-0.128*** (0.004)
Sample Size	7,585,979	5,578,631	5,105,431	5,105,431
Sample Restriction	-	Econ. Active	Worked	Worked
Mean (Without Disabilities)	0.788	0.068	0.184	7.438
Mean (Severe Difficulty)	0.553	0.157	0.255	7.029
Mean (Some Difficulty)	0.700	0.131	0.237	7.127
Individual Controls	✓	✓	✓	✓
Occup. & Sector Controls	×	✓	✓	✓

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table displays gaps in labor market outcomes across disability groups using the 2010 Census. The sample includes working-age individuals aged 25 to 54. To capture severe disabilities, I construct an indicator variable for whether individuals report having a permanent intellectual disability or having severe difficulties in one or more of the following activities: seeing, hearing, walking, or climbing stairs. To capture some disabilities, I create an indicator variable for whether individuals report having some difficulty in hearing, walking, or climbing stairs. I regress labor market outcomes on indicators for severe and some difficulties, with the omitted group being individuals with no disabilities (no difficulties). All specifications include potential experience, potential experience squared, indicators for educational categories and rural residence, and municipality fixed effects. Columns (2)–(4) additionally include occupation and economic-sector fixed effects. The dependent variables are indicators for being economically active, unemployment in the last week of July of 2010, employment in the informal sector, and the inverse hyperbolic sine transformation of work income. Column (2) further restricts the sample to economically active individuals. In Columns (3) and (4), the sample is restricted to individuals who worked in the last week of July 2010. Means of dependent variables across disability groups are reported. Standard errors are clustered at the state level.

A.2 List of Disabilities

As explained in Section 2.2, the Decree 3,298, Article 4, provided the set of disabilities that qualify for reserved jobs under the AA quota regulation. Over time, this list has been expanded to incorporate additional conditions through decisions by the Ministry of Labor, Labor Courts, and the Supreme Court. Examples of qualifying categories include:

Physical Disabilities. Complete or partial alteration of one or more segments of the human body that impairs physical functioning. Examples include paraplegia, paraparesis, monoplegia, monoparesis, tetraplegia, tetraparesis, triplegia, triparesis, hemiplegia, hemiparesis, ostomy, amputation or absence of a limb, cerebral palsy, dwarfism, and congenital or acquired limb deformities (excluding purely aesthetic deformities or those without functional impact). Individuals with reduced mobility may also qualify.

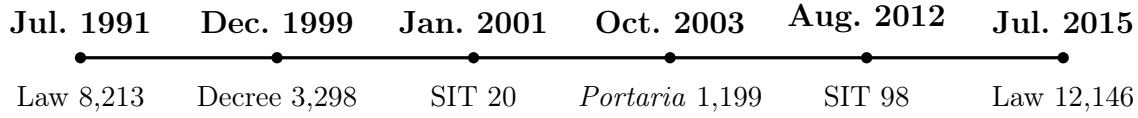
Hearing Loss. Bilateral (partial or total) loss of 41 decibels (dB) or more, measured by an audiogram at the frequencies 500 Hz, 1,000 Hz, 2,000 Hz, and 3,000 Hz. This corresponds to moderate to profound hearing loss.

Blindness and Vision Impairment. Blindness is defined as visual acuity equal to or less than 0.05 in the better eye with best optical correction. Low vision refers to visual acuity between 0.3 and 0.05 in the better eye with best correction. Visual impairment also includes cases in which the combined visual field in both eyes is 60 degrees or less, as well as the simultaneous occurrence of any of these conditions. Monocular vision, which is visual acuity equal to or less than 0.05 in one eye with best optical correction, was added to the list in 2011 (CONJUR/MTE 444/11).

Cognitive Disorders. Permanent cognitive or mental impairments that limit functioning in two or more of the following domains: communication, personal care, social skills, use of community resources, health and safety, academic skills, leisure, and work. Examples include learning disabilities such as dyscalculia and autism spectrum disorder. Autism spectrum disorder was added to the list in 2012 (Law 12,764).

Multiple Disabilities. Defined as the presence of two or more of the qualifying disabilities listed above.

Figure A1: Timeline of Key Regulations Related to the AA Quotas



A.3 Timeline of Disability Labor Market Regulations

July 1991. The AA quota originated in 1991 with Law 8,213, enacted as part of a social security reform. In the absence of clear definitions of qualifying disabilities or explicit procedures for monitoring and enforcing compliance, the law was largely ineffective in practice (Costilla et al., 2002).

December 1999. The National Policy for People with Disabilities (Decree 3,298) is enacted. The new regulation consisted of a set of guidelines for ensuring the rights and integration of people with disabilities into society. Relevant to this paper, Chapter I and Section IV defined a list of disabilities (see Appendix A.2) and assigned to the Ministry of Labor the responsibility of oversight and enforcement, through labor inspections, of the AA quota.

January 2001. The Ministry of Labor issued Normative Instruction SIT 20, which for the first time mandated that all labor inspections include an assessment of compliance with the AA quota. The instruction also set out the criteria inspectors should use to determine which firms are subject to the quota.

October 2003. To strengthen enforcement of the AA quota, the Ministry of Labor issued Portaria 1,199, which for the first time established administrative fines and penalties for non-compliance.

August 2012. The Ministry of Labor issued Normative Instruction SIT 98, creating the possibility of special, targeted inspections specifically aimed at assessing compliance with the AA quota.

July 2015. The federal government enacted Law 13,146, the Brazilian Inclusion Law (Statute of Persons with Disabilities), which established a broad set of rights for people with disabilities and redefined disability using a social, rather than medical, framework. Article 5 of the statute also affirms that “persons with disabilities shall be protected from all forms of neglect, discrimination, exploitation, violence, torture, cruelty, oppression, and inhuman or degrading treatment.” The Law entered into force in January 2016.

A.4 Data Appendix

The Labor Market Data. The *Relação Anual de Informações Sociais* (RAIS) is the main labor market data source used in this paper. RAIS is a linked employer-employee administrative registry containing unique tax identifiers for workers and firms, which allows me to construct both firm- and worker-level samples. The raw data are provided in state-year files, and all variables are standardized across years. I use data from 2003, the first year in which employers reported disability status, until 2018. Information on disability type has been available since 2006 and includes six categories: physical, hearing, visual, intellectual, multiple, and rehabilitated.

The Census Data. The Brazilian Demographic Census, administered decennially, offers comprehensive population coverage, including individuals not captured in the formal labor market. For this paper, I use the 2000 and 2010 editions. Changes in the disability questionnaire render the 2022 Census incomparable with previous editions.

Disability. Information on disabilities was first collected in the 1991 Census. Individuals who identified as having a disability were asked whether they had one of the following conditions: (i) blindness; (ii) deafness; (iii) hemiplegia (paralysis on one side of the body); (iv) paraplegia (paralysis of the legs); (v) quadriplegia (total paralysis); (vi) absence of one or more limbs, either congenital or due to illness or accident; (vii) intellectual disability since childhood; (viii) multiple disabilities (two or more of the previous conditions); or (ix) none of the above. Only 1.23 percent of the working-age population reported having a disability.

Following methodologies developed by the Washington Group on Disability Statistics, the 2000 and 2010 Census editions introduced a new set of disability questions. These Censuses asked four questions: (i) whether the respondent has a permanent intellectual disability that limits regular activities, such as working or attending school; (ii) how difficult it is for the respondent to see; (iii) how difficult it is for the respondent to hear; and (iv) how difficult it is for the respondent to walk or climb stairs. For questions (ii)–(iv), individuals could answer: unable to do the activity, very difficult, some difficulty, or no difficulty.³³

In the 2000 and 2010 Censuses, 12.45 percent and 17.23 percent of working-age individuals reported having “some difficulty” with at least one activity, while 3.90 percent and

³³Although the 2000 Census also asked whether respondents had total and permanent paralysis (quadriplegia), paralysis of the legs (paraplegia), paralysis on one side of the body (hemiplegia), or absence of one or more limbs, these questions were removed in the 2010 Census. I therefore do not use this information in the empirical analysis in Section 4.

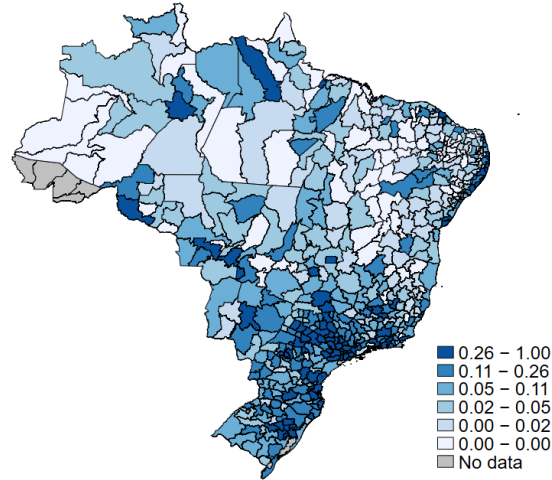
6.68 percent reported being unable to perform the activity or finding it very difficult. An alternative measure that excludes seeing from the list of “some difficulty” activities yields disability rates of 4.48 percent in 2000 and 6.11 percent in 2010. Given substantial differences in measurement and incidence, my empirical analysis excludes the 1991 Census.

Labor Market Information. To ensure comparability of labor market information across the 2000 and 2010 Census editions, I focus on questions that are asked consistently in both editions. For the analysis in Tables 1 and 2, I create the following variables for individuals with and without disabilities: (i) formal employment; (ii) informal employment; (iii) work income (earnings from main and secondary jobs); and (iv) non-work income (including retirement and pension benefits, rental income, alimony, donations, welfare programs, unemployment insurance, and other sources). For the hours analysis in Table C3, Appendix C, I use two measures: (i) conditional hours, defined as weekly hours worked at the main job; and (ii) unconditional hours, which assigns zero hours to individuals who do not report working any hours.

Other Information. Table C3, Appendix C, also considers two additional outcomes: (i) migration, an indicator for individuals who report having moved to another state in the past five years; and (ii) welfare program participation, an indicator for individuals who report receiving welfare benefits from the government.

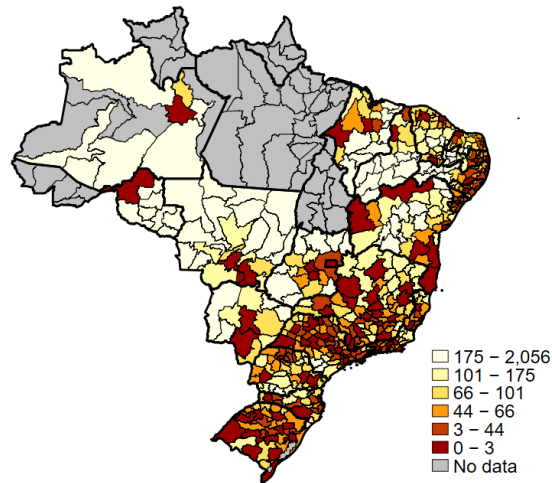
B Figures (Aggregate Analysis)

Figure B1: Cross-Market Variation: Potential Demand



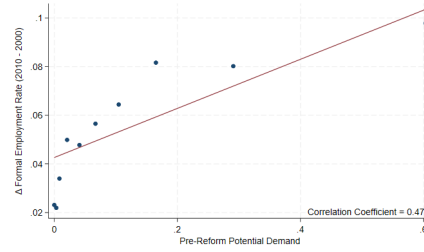
Note: The graph illustrates the geographic variation in the potential-demand measure across local labor markets (micro-regions) in Brazil. Potential demand is defined as the total number of private-sector jobs that would be available under the hiring-quota regulation in 1998, calculated from the distribution of firm size and divided by the total number of people with disabilities in 2000. Sources: 1998 RAIS and 2000 Demographic Census data.

Figure B2: Cross-Market Variation: Enforcement Capacity



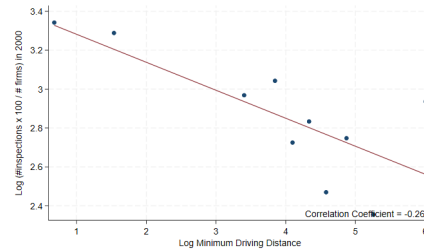
Note: The graph illustrates the geographic variation in enforcement capacity across local labor markets (micro-regions) in Brazil. Enforcement capacity is computed by measuring the driving distance between the centroid of each municipality and the nearest labor office created prior to the 2000s. I then define the enforcement-capacity measure for each micro-region as the minimum of these distances across its municipalities. Sources: Data on labor offices from the Ministry of Labor and from [Ponczek and Ulyssea \(2021\)](#).

Figure B3: Pre-Reform Potential Demand and Actual Employment Rate



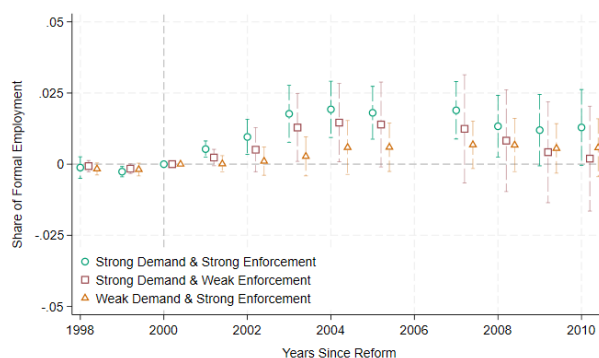
Note: This binned scatter plot shows the relationship between pre-reform potential demand and the increase in the formal-sector employment rate for people with disabilities between 2000 and 2010. Potential demand is defined as the total number of private-sector jobs that would be available under the hiring-quota regulation in 1998, calculated from the distribution of firm size and divided by the total number of people with disabilities in 2000. The increase in the employment rate is defined as the share of formal employment in 2010 relative to 2000. The unit of observation is the micro-region. The right-hand-side variable is grouped into 10 bins. The correlation coefficient is 0.470. Sources: 1998 RAIS and 2000 and 2010 Demographic Census data.

Figure B4: Pre-Reform Driving Distance to Labor Offices and Frequency of Inspections



Note: This binned scatter plot shows the relationship between the pre-reform distance to the nearest labor office and the frequency of labor inspections in 2000. The former is defined as the minimum driving distance between each micro-region and the nearest labor office. The latter is defined as the log of the total number of inspections normalized by the number of firms in each micro-region in 2000. The unit of observation is the micro-region. The right-hand-side variable is grouped into 10 bins. The correlation coefficient is -0.262 . Sources: Data on labor offices (from the Ministry of Labor and [Ponczek and Ulyssea \(2021\)](#)) and data on labor inspections.

Figure B5: Impact of the 2000 Reform on Formal Employment of People with Disabilities



Note: This figure reports point estimates of the annual effects of the 2000 reform on the share of formal employment among people with disabilities, using both RAIS and Census data. The numerator and denominator are constructed from RAIS records (excluding 2006 due to reporting inconsistencies) and Census records. The omitted category is micro-regions with weak enforcement and weak demand in 2000.

C Tables (Aggregate Analysis)

Table C1: Descriptive Statistics at the Micro Region Level

	(1) Mean	(2) SD
Panel A: People Without Disabilities		
Formal Employment	0.192	0.100
Informal Employment	0.205	0.049
Work Income (in BRL <i>reais</i>)	638.17	316.59
Non-Work Income (in BRL <i>reais</i>)	70.24	39.03
Panel B: People With Disabilities		
Formal Employment	0.117	0.065
Informal Employment	0.148	0.041
Work Income (in BRL <i>reais</i>)	367.17	235.54
Non-Work Income (in BRL <i>reais</i>)	119.88	53.89
Panel C: Demographic and Economic Variables		
Share Female	0.498	0.012
Literacy Rate	0.707	0.113
Share College Educated	0.018	0.016
Share Urban	0.677	0.180
Unemployment Rate	0.154	0.114
Population	304,299.59	799,492.16
Number of Micro Regions	558	

Note: This table reports descriptive statistics (means and standard deviations) for the year 2000 at the micro-region level. Shares of formal and informal employment, average work and non-work incomes, shares of female, college-educated, and urban populations, the unemployment rate, average income per capita, and average population are computed using individual-level data and sampling weights from the 2000 Census data.

Table C2: Aggregate Analysis: Impacts on Formal and Informal Work Income

	(1) With Disabilities Formal	(2) Informal	(3) Without Disabilities Formal	(4) Without Disabilities Informal
(Strong Demand & Strong Enforcement) $\times$ Reform	10.188* (5.221)	-5.014 (3.093)	7.523 (6.725)	-5.014 (3.093)
(Strong Demand & Weak Enforcement) $\times$ Reform	9.756 (7.510)	-6.632 (4.860)	11.191 (7.249)	-6.632 (4.860)
(Weak Demand & Strong Enforcement) $\times$ Reform	1.619 (4.516)	-6.333* (3.235)	8.093 (6.289)	-6.333* (3.235)
Sample Size	1,116	1,116	1,116	1,116
Mean Dep. Var. (in 2000)	103.55	81.83	192.77	143.53
Mean Dep. Var.	135.94	75.41	247.29	119.59

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the aggregate effects of reform on formal and informal work income for people with disabilities (Columns (1) and (2)) and people without disabilities (Columns (3) and (4)). All columns refer to Equation (2). Means of dependent variables are computed from all micro-regions. Standard errors are clustered at the micro-region level.

Table C3: Impacts on Migration, Hours Worked, and Welfare Program Take-Up

	(1) Interstate Migration	(2) Conditional Hours	(3) Unconditional Hours	(4) Welfare Program
(Strong Demand & Strong Enforcement) $\times$ Reform	0.001 (0.002)	0.259 (0.286)	0.594** (0.267)	-0.021*** (0.004)
(Strong Demand & Weak Enforcement) $\times$ Reform	0.001 (0.002)	0.229 (0.289)	-0.153 (0.307)	-0.007 (0.004)
(Weak Demand & Strong Enforcement) $\times$ Reform	0.000 (0.002)	0.357 (0.317)	-0.057 (0.277)	-0.011** (0.005)
Sample Size	1,116	1,116	1,116	1,116
Mean Dep. Var. (in 2000)	0.031	43.31	21.46	0.012
Mean Dep. Var.	0.028	41.30	21.84	0.095

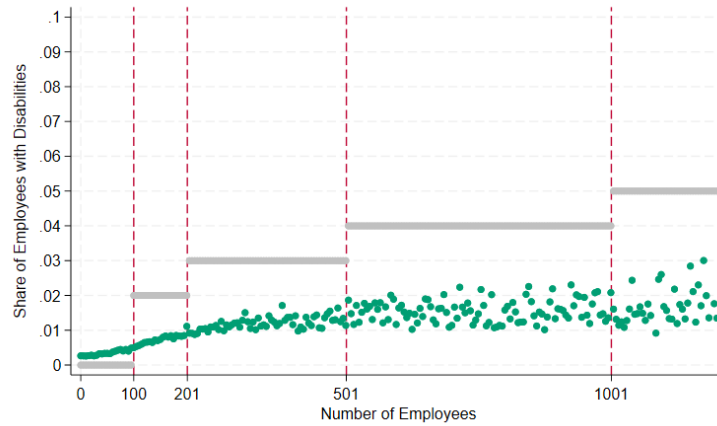
Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the aggregate effects of reform on the interstate migration rate, the (conditional and unconditional) number of hours workers, and the likelihood of receiving a welfare program for people with disabilities. Migration is defined as having moved to another state in the last five years. Conditional hours correspond to the weekly hours worked at the main job, whereas unconditional hours assign zero hours to individuals who do not report any weekly hours worked. All columns refer to Equation (2). Means of dependent variables are computed from all micro-regions. Standard errors are clustered at the micro-region level.

D Firm-Level Analysis: Alternative Empirical Strategies

D.1 Regression Discontinuity Design

The institutional context described in Section 2.2 suggests that a natural empirical strategy to estimate the causal impact of disability hiring quotas would be a regression discontinuity design. Private-sector firms with at least 100 employees must reserve a minimum of 2 percent of positions for people with disabilities or individuals enrolled in a vocational rehabilitation program. The required share increases with firm size: firms with 201–500 employees must meet a 3 percent quota, those with 501–1,000 employees must meet a 4 percent quota, and firms with more than 1,000 employees face a 5 percent quota. However, the data show that firms do not comply with these hiring quotas in the absence of labor inspections. Figure D1 documents no visible discontinuities in the share of employees with disabilities around the statutory cutoff points.

Figure D1: No First-Stage



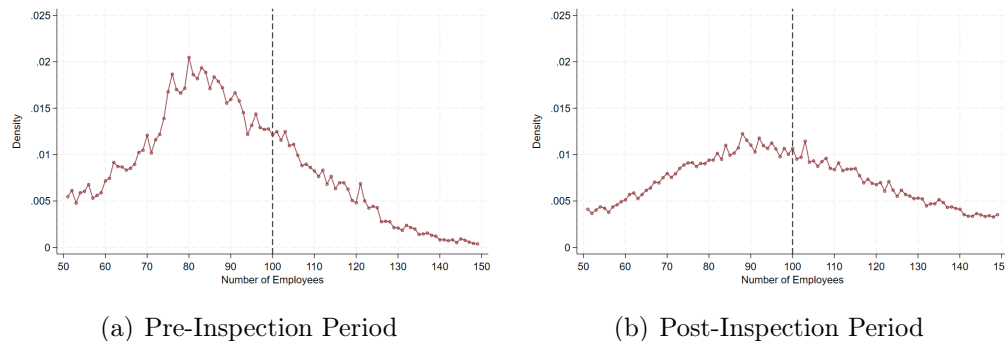
Note: This figure reports the share of employees with disabilities across the running variable using RAIS data from 2003 to 2011. The running variable is defined as the number of employees at the firm by the end of each year. Red dashed lines represent the cutoff requirements. Grey horizontal lines display the counterfactual share of employees with disabilities in case of full compliance.

D.2 Bunching

Another potential empirical strategy is the bunching method. If firms perceived disability hiring quotas as a costly regulation, they might respond by adjusting employment to remain just below the thresholds. Figure D2 shows no evidence of such endogenous responses: firms

do not bunch around the quota thresholds, either before or after labor inspections. This pattern reinforces the interpretation that the 2 percent hiring requirement does not impose detectable costs on firms and rules out the feasibility of bunching-based approaches.

Figure D2: No Bunching



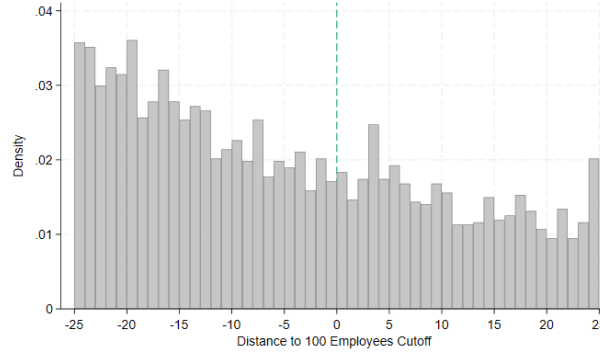
Note: Figures display the distribution of workers for all firms in the estimation sample before and after the labor inspections, using a bin size of one. Sources: RAIS and data on inspections.

D.3 Other Cutoffs

Section 5 focuses on the *extensive* margin of modest quotas by comparing firms just above the 100-employee threshold—those subject to the 2 percent disability hiring quota—to firms just below it, which are not required to hire workers with disabilities. The estimates therefore capture the effects of enforcing a modest quota at its extensive margin. The institutional setting, however, also creates opportunities to study *intensive* margins of the policy by exploiting the additional quota thresholds at 200, 500, and 1,000 employees. In practice, these thresholds yield very small samples: only 774 firms fall between 150 and 250 employees, 50 firms between 450 and 550, and just 6 firms between 950 and 1,050 prior to inspection. I pool these groups and estimate an alternative version of Equation (3), in which $Quota_{j,pre}$ is an indicator for firms above 200, 500, or 1,000 employees, and the post-event coefficients β_k^{Quota} measure the dynamic effects of an additional 1 percentage point quota requirement. While estimates move in the same direction as the main results, the extremely small samples at these thresholds yield imprecise estimates with little statistical power.

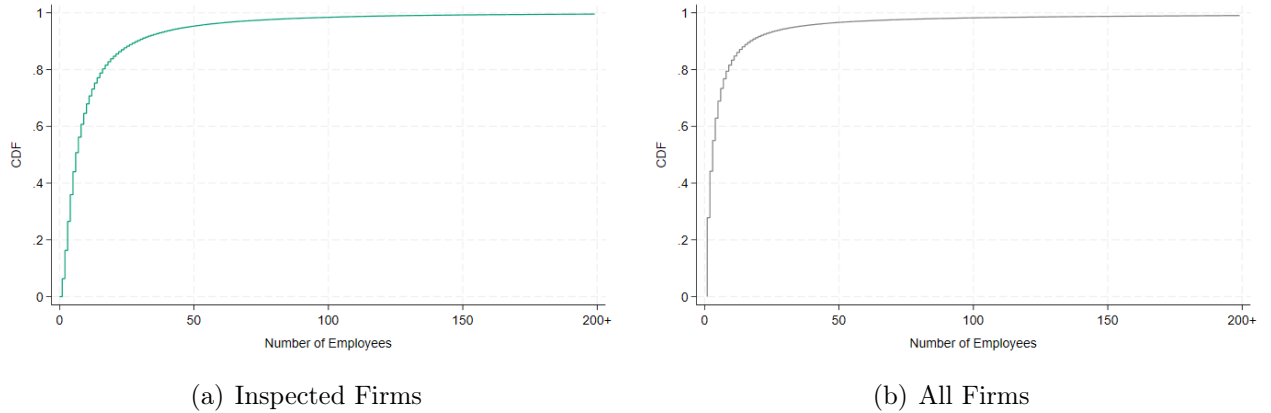
E Figures (Firm- and Worker-Level Analyses)

Figure E1: Histogram



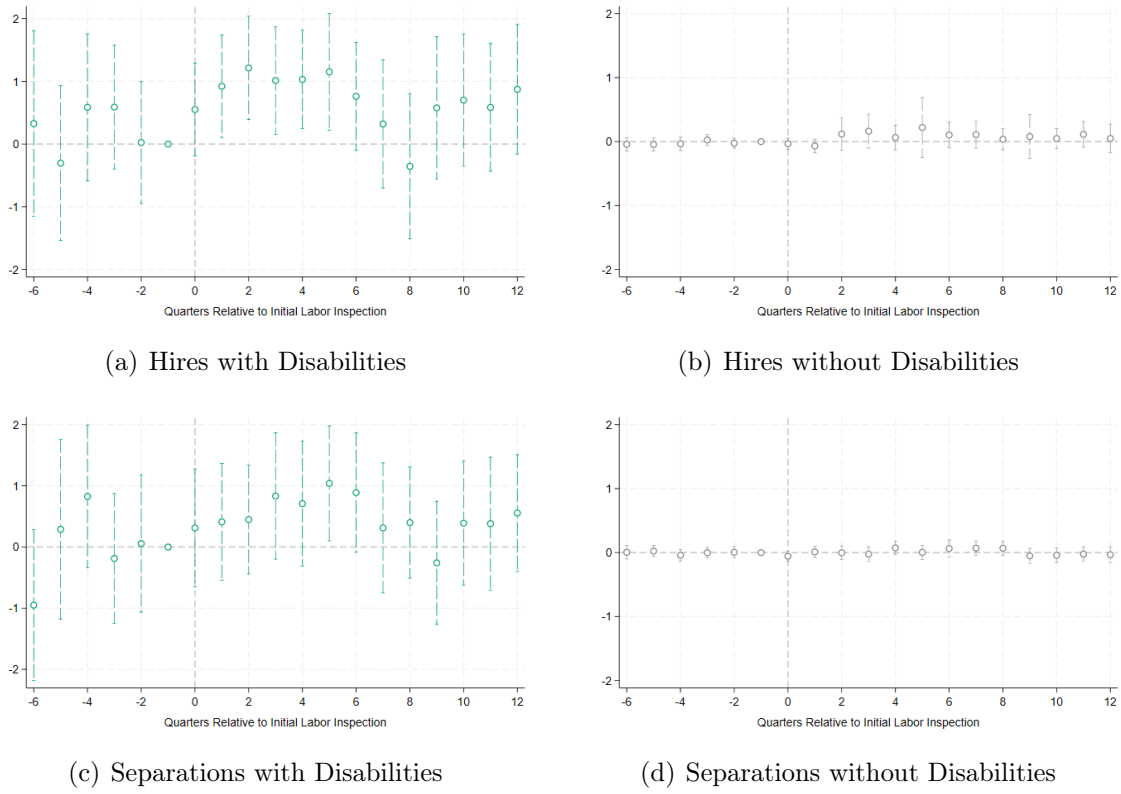
Note: Histogram depicts the distribution of firm size, measured by total number of employees, relative to the AA quota cutoff of 100 employees within one point bins. Further details about the sample can be found in Table F1. Sources: RAIS and data on inspections.

Figure E2: Cumulative Distribution Function of Firm Size



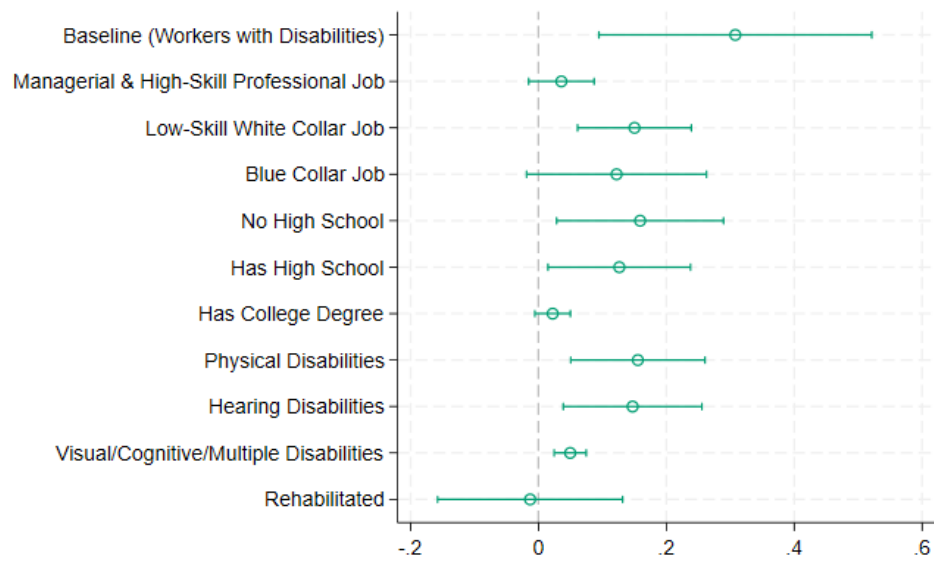
Note: This figure reports cumulative distribution functions of firm size. On the right, I consider the sample of inspected firms. On the left, I consider the universe of formal firms in 2003. Sources: RAIS and data on inspections.

Figure E3: Effects of AA Quotas on New Hires and Separations



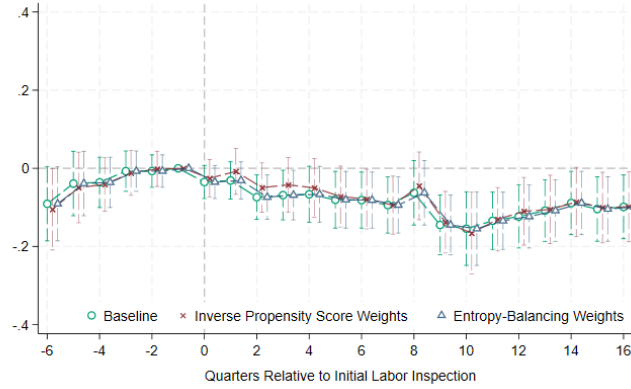
Note: This figure reports point estimates of the quarterly effects of AA quota on hires and separations of workers with and without disabilities. All graphs refer to a Poisson model with year and firm fixed effects and without time-varying firm-level controls.

Figure E4: Heterogeneous Employment Effects of AA Quotas

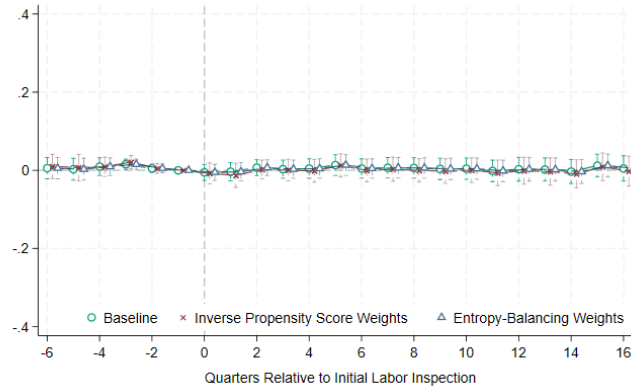


Note: This figure reports aggregate point estimates of the aggregate effects of AA quota on employment of people with disabilities considering heterogeneity across jobs, educational levels, and disabilities. The baseline specification refers to Column (1) of Table 3.

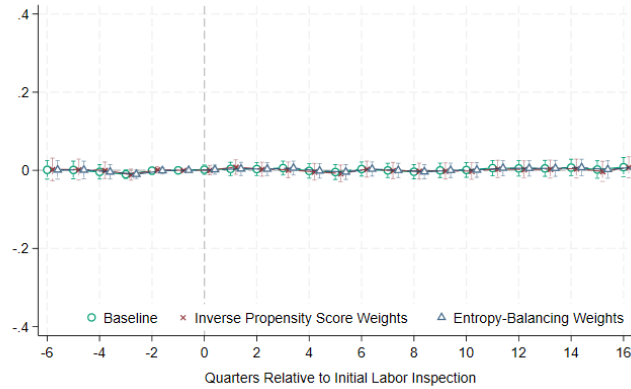
Figure E5: Robustness of Selected Worker-Level Effects: Re-Weighting Methods



(a) Log Hourly Wages



(b) Log Number of Contracted Hours



(c) Part-Time Employment

Note: This figure reports point estimates of the quarterly effects of AA quota on log hourly wages, log number of contracted hours, and the likelihood of part-time employment for workers with disabilities using re-weighting methods. The green circle estimates repeat the baseline specification from Figure 4(a). The red x estimates refer to coefficients using inverse propensity score weights from the following pre-inspection characteristics: gender, race, age, squared age, education, and occupation. The blue triangle estimates show coefficients using entropy-balancing weights from [Hainmueller \(2012\)](#) with the same set of characteristics.

F Tables (Firm- and Worker-Level Analyses)

Table F1: Descriptive Statistics: Firm-Level Sample

	(1)		(2)		(3)		(4)	
	Control Firms		Treated Firms					
	Pre		Post		Pre		Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Main Variables								
# Employees without Disabilities	74.85	26.78	96.87	81.96	96.24	36.88	130.02	229.24
# Employees with Disabilities	0.31	3.63	0.47	4.16	0.29	2.56	0.77	4.30
Has at Least One Employee with Disabilities	0.09	0.28	0.14	0.35	0.13	0.33	0.26	0.44
Share Employees with Disabilities (x100)	0.38	4.41	0.43	3.68	0.31	2.74	0.59	3.73
Average Earnings	2,089.29	1,837.75	2,193.67	1,859.65	2,105.76	1,935.02	2,237.90	1,931.06
Log Average Earnings	7.48	0.52	7.52	0.62	7.47	0.53	7.51	0.73
Location								
Central West Region	0.07	0.26	0.07	0.26	0.05	0.22	0.05	0.22
North Region	0.04	0.20	0.04	0.20	0.04	0.20	0.04	0.20
Northeast Region	0.11	0.31	0.11	0.31	0.11	0.31	0.11	0.31
South Region	0.22	0.42	0.22	0.41	0.26	0.44	0.26	0.44
Southeast Region	0.56	0.50	0.56	0.50	0.54	0.50	0.54	0.50
Economic Sector								
Construction	0.07	0.25	0.07	0.26	0.07	0.26	0.08	0.27
Commerce	0.20	0.40	0.19	0.39	0.17	0.37	0.16	0.37
Transp., Storage & Commun.	0.09	0.28	0.09	0.28	0.08	0.28	0.08	0.28
Transformation Industry	0.29	0.45	0.30	0.46	0.28	0.45	0.29	0.45
Real Estate	0.15	0.36	0.15	0.36	0.18	0.38	0.17	0.38
Services	0.04	0.19	0.04	0.19	0.04	0.19	0.04	0.19
Other Categories	0.17	0.37	0.17	0.37	0.18	0.38	0.18	0.38
N	2,073		2,073		1,199		1,199	

Note: This table reports descriptive statistics for firms using information from RAIS data. The first two columns refer to a sample of control firms. Columns (3) and (4) report summary statistics for the treatment group. Further details on the sample construction are found in Section 5. Odd columns indicate summary statistics using the averages before labor inspection while even columns refer to the period following it. The variables are total number of employees without and with disabilities, indicator for having at least one employee with disabilities, share of employees with disabilities, log average earnings, indicator variables for whether the establishment is located in Central-West, North, Northeast, South and Southeast regions, average population of the municipality in which the establishment is located, and indicator variables for economic sector the establishment belongs to (construction, commerce, transportation, storage and communication, transformation industry, real estate, services, or other sectors). Sources: RAIS and data on inspections.

Table F2: Descriptive Statistics of Labor Violations

	(1)		(2)		(3)	
	Control Firms		Treated Firms		Full Sample	
	Mean	SD	Mean	SD	Mean	SD
Inspection Year	2,008.44	2.26	2,008.45	2.26	2,004.53	3.60
Formal Registration	0.42	0.49	0.43	0.49	0.34	0.47
Working Hours	0.19	0.39	0.16	0.37	0.10	0.31
Rest	0.14	0.34	0.12	0.33	0.08	0.28
Wages	0.14	0.35	0.13	0.33	0.13	0.33
Severance Pay	0.21	0.41	0.19	0.39	0.32	0.47
Unemployment Insurance	0.01	0.12	0.01	0.09	0.01	0.11
RAIS	0.04	0.20	0.03	0.18	0.03	0.18
Transportation Subsidy	0.03	0.16	0.03	0.18	0.04	0.19
Other Violations	0.34	0.48	0.34	0.47	0.34	0.47

Note: This table reports descriptive statistics of labor violations. The first two columns refer to a sample of control firms, whereas Columns (3) and (4) report summary statistics for the treatment group. The last two columns correspond to the full sample of labor inspections. Further details on the sample construction are found in Section 5. I compute statistics for inspection timing and violations found during inspections, including those related to formal registration, working hours, rights to rest, wages, severance pay, unemployment insurance, RAIS, transportation subsidies, and other labor violations. Sources: RAIS and data on labor inspections.

Table F3: Correlation Between First Inspection Timing and Firm Characteristics

	(1) 2005	(2) 2006	(3) 2007	(4) 2008	(5) 2009	(6) 2010	(7) 2011	(8) 2012
# Employees with disabilities	0.001 (0.002)	0.003 (0.002)	-0.004* (0.002)	-0.004* (0.002)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
# Employees without disabilities	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)	-0.000 (0.000)
Location: North Region	0.014 (0.033)	0.046 (0.035)	-0.047 (0.030)	-0.046 (0.033)	-0.037 (0.033)	0.062 (0.039)	0.039 (0.045)	-0.031 (0.032)
Location: Northeast region	-0.013 (0.025)	-0.007 (0.025)	-0.006 (0.027)	-0.019 (0.029)	0.022 (0.029)	0.020 (0.029)	0.009 (0.035)	-0.007 (0.026)
Location: Southeast Region	0.030 (0.021)	0.025 (0.021)	0.008 (0.022)	-0.008 (0.024)	0.015 (0.024)	0.016 (0.024)	-0.062* (0.029)	-0.024 (0.022)
Location: South Region	0.020 (0.023)	0.027 (0.023)	0.016 (0.024)	-0.013 (0.026)	0.008 (0.026)	0.007 (0.025)	-0.052 (0.030)	-0.014 (0.023)
Sector: Transformation Industry	-0.064 (0.036)	0.006 (0.038)	0.014 (0.037)	0.006 (0.033)	0.014 (0.039)	-0.005 (0.039)	0.006 (0.039)	0.022 (0.028)
Sector: Construction	-0.116*** (0.027)	-0.049 (0.027)	-0.011 (0.028)	0.081** (0.029)	0.016 (0.031)	-0.020 (0.030)	0.019 (0.031)	0.080** (0.025)
Sector: Commerce	-0.050 (0.027)	-0.036 (0.026)	-0.011 (0.025)	0.025 (0.024)	0.019 (0.028)	-0.022 (0.027)	0.047 (0.029)	0.028 (0.020)
Sector: Transportation and Storage	-0.070* (0.030)	-0.045 (0.029)	-0.026 (0.028)	0.040 (0.028)	0.093** (0.035)	0.020 (0.033)	-0.023 (0.030)	0.011 (0.022)
Sector: Real Estate	-0.069** (0.026)	-0.020 (0.026)	-0.025 (0.024)	0.042 (0.024)	-0.023 (0.026)	0.006 (0.027)	0.038 (0.028)	0.052* (0.020)
Sector: Others	-0.053* (0.023)	-0.012 (0.023)	-0.013 (0.021)	0.036 (0.019)	-0.007 (0.023)	-0.001 (0.024)	0.027 (0.023)	0.024 (0.016)
Average Earnings	-0.000*** (0.000)	-0.000* (0.000)	-0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)
Share of Employees with Disabilities	-0.001 (0.002)	-0.000 (0.002)	0.005* (0.002)	0.005* (0.002)	-0.001 (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.001 (0.001)
Wage Bill	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000** (0.000)	0.000*** (0.000)	-0.000 (0.000)	-0.000* (0.000)

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the correlation between the timing of the first inspection and firm characteristics by regressing indicator variables for the year of the first inspection on firm characteristics, controlling for calendar-month-of-inspection fixed effects. Each column corresponds to an indicator for a first-inspection year from 2005 to 2012. Firm characteristics include the number of employees with and without disabilities; indicator variables for whether the firm is located in the Central-West, North, Northeast, South, or Southeast regions; indicator variables for the firm's economic sector; average earnings; the share of employees with disabilities; and the wage bill. Standard errors are clustered at the firm level. Sample size is 64,182. Sources: RAIS and inspection records.

Table F4: Robustness Checks: Firms-Level Results

	(1) employm.	(2) employm. (relabeling)	(3) (IHS) employm.	(4) employm.	(5) employm. (non-disabled)	(6) employm.	(7) employm.	(8) employm.	(9) employm.	(10) employm.
Panel A: Dynamic Impacts										
Immediate ($k = 0$)	-0.037 (0.080)	0.004 (0.020)	0.025** (0.010)	-0.072 (0.195)	-0.003 (0.008)	-0.034 (0.080)	-0.061 (0.095)	-0.049 (0.113)	0.048 (0.048)	0.044 (0.064)
Short Run ($k = 6$)	0.413** (0.195)	0.231** (0.096)	0.121*** (0.023)	0.505 (0.359)	0.040 (0.052)	0.397** (0.193)	0.464** (0.229)	0.413* (0.243)	0.627*** (0.240)	0.414* (0.236)
Long Run ($k = 12$)	0.426*** (0.152)	0.198*** (0.076)	0.161*** (0.027)	0.449 (0.370)	0.012 (0.046)	0.409*** (0.154)	0.464** (0.184)	0.405* (0.217)	0.529*** (0.167)	0.427*** (0.123)
Panel B: Aggregate Impacts										
Post $\times$ Treated	0.308*** (0.109)	0.162*** (0.052)	0.131*** (0.018)	0.480** (0.244)	0.041 (0.046)	0.311*** (0.115)	0.276** (0.130)	0.267** (0.136)	0.444*** (0.133)	0.350*** (0.115)
Sample Size	65,440	65,440	65,440	65,440	65,440	65,440	51,080	37,299	52,020	40,040
Firm and Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
State and Industry Trends	✓	✓	✓	×	×	×	✓	✓	✓	✓
Sample Restriction	75-125 emp.	75-125 emp.	75-125 emp.	75-125 emp.	75-125 emp.	75-125 emp.	80-120 emp.	85-115 emp.	75-125 emp.	75-125 emp.
Model	OLS	OLS	OLS	Poisson	Poisson	OLS	OLS	OLS	OLS	OLS
# Firms	3,272	3,272	3,272	3,272	3,272	3,272	2,554	1,865	2,601	2,002
Mean Dep. Var (Control)	0.310	0.112	0.108	0.310	74.854	0.310	0.306	0.337	0.302	0.255

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports several robustness checks for the firm-level analysis. Column (1) repeats Column (1) from Table 3. Column (2) refers to the relabeling of existing workers with disabilities as the dependent variable. Column (3) refers to the inverse hyperbolic sine transformation of the number of workers with disabilities as the dependent variable. Columns (4) and (5) estimate a conditional fixed-effect Poisson model using the number of employees with and without disabilities as dependent variables (without state- and industry-specific trends). Column (6) excludes state- and industry-specific trends from the set of firm controls. Columns (7) and (8) consider narrower windows around the cutoff. Columns (9) and (10) exclude firms very close to the cutoff (by excluding 5 and 10 employees window). Means of dependent variables are computed from the control group in the quarterly window $[-6, -1]$ before the labor inspection. Standard errors are clustered at the firm level.

Table F5: Descriptive Statistics: Worker-Level Sample

	(1)		(2)		(3)		(4)	
	Control Workers		Treated Workers					
	Before		After		Before		After	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Main Variables								
Has Disability	0.00	0.06	0.01	0.07	0.00	0.05	0.01	0.08
Hourly Earnings	12.72	29.75	14.55	32.57	12.42	24.28	14.20	27.42
Log Hourly Earnings	2.28	0.66	2.38	0.71	2.28	0.66	2.37	0.71
Contracted Hours (Monthly)	185.43	23.78	184.55	24.47	184.75	23.60	183.79	24.86
Male	0.66	0.47	0.66	0.48	0.65	0.48	0.63	0.48
White	0.66	0.47	0.61	0.49	0.68	0.47	0.63	0.48
Has College Degree	0.10	0.30	0.11	0.31	0.10	0.30	0.11	0.32
Manager	0.03	0.18	0.03	0.17	0.03	0.17	0.03	0.17
Professional	0.15	0.35	0.15	0.36	0.15	0.36	0.16	0.37
Low-Skill White Collar	0.16	0.37	0.17	0.38	0.16	0.36	0.17	0.38
Low-Skill Blue Collar	0.66	0.47	0.65	0.48	0.66	0.47	0.63	0.48
Location								
Central West Region	0.07	0.26	0.07	0.26	0.05	0.21	0.05	0.22
North Region	0.04	0.20	0.05	0.21	0.04	0.19	0.04	0.21
Northeast Region	0.10	0.30	0.11	0.31	0.11	0.31	0.11	0.31
South Region	0.22	0.42	0.21	0.41	0.27	0.44	0.24	0.42
Southeast Region	0.56	0.50	0.56	0.50	0.54	0.50	0.55	0.50
Sector								
Construction	0.06	0.24	0.08	0.27	0.06	0.25	0.08	0.26
Commerce	0.19	0.40	0.18	0.39	0.16	0.37	0.15	0.36
Transp., Storage & Commun.	0.08	0.28	0.09	0.28	0.08	0.27	0.08	0.28
Transformation Industry	0.29	0.45	0.29	0.45	0.28	0.45	0.26	0.44
Real Estate	0.15	0.36	0.16	0.37	0.18	0.38	0.22	0.42
Services	0.04	0.20	0.04	0.20	0.04	0.20	0.04	0.20
Other Categories	0.18	0.38	0.16	0.37	0.19	0.39	0.17	0.38
N	842,569				709,715			

Note: This table reports descriptive statistics for workers using information from RAIS data. The first two columns refer to a sample of workers from control firms. Columns (3) and (4) report summary statistics for the treatment workers. Further details on the sample construction are found in Section 5. Odd columns indicate summary statistics using the averages in the quarterly window [-6,-1] before labor inspection while even columns refer to the quarterly window [0, 12] following it. The variables are: an indicator for having a disability, hourly wages, log hourly wages monthly number of hours contracted, indicators for male, white, having a college degree, indicators for the manager, professional, low-skill white collar, and blue-collar positions, indicator variables for whether the establishment is located in Central-West, North, Northeast, South, and Southeast regions, and indicator variables for economic sector the establishment belongs to (construction, commerce, transportation, storage and communication, transformation industry, real estate, services, or other sectors).

G HR Survey: Overview and Results

G.1 Survey Overview

To understand how people with disabilities are perceived in the labor market, I designed and implemented a novel employer survey in collaboration with Oppen Social. The survey instrument included a vignette experiment, questions about support for affirmative action, and questions regarding challenges and concerns about hiring workers with disabilities.

The final sample consists of 60 firms identified in the RAIS data, 31 of which have more than 100 employees. Because of substantial differences between very small and larger firms, I exclude firms with fewer than 50 employees from the initial sampling frame. In June 2022, Oppen Social attempted, over a seven-day period, to contact HR representatives from 764 firms with fewer than 100 employees and 751 firms with 100 or more employees, inviting them to complete a 30-minute phone interview. When a firm lacked an HR department, the interview was conducted with an employee familiar with its hiring process. The response rate was 4 percent, which is consistent with previous surveys conducted by Oppen Social and typical for lengthy phone surveys targeting HR personnel.

Around 52 percent of surveyed firms are located in the Southeast region, 19 percent in the Northeast, 17 percent in the South, 9 percent in the Central-West, and 3 percent in the North. In terms of industry composition, 18 percent of firms operate in manufacturing, 4 percent in transportation, and 16 percent in wholesale or retail. The distribution of respondents across industries and regions is similar to the national firm population (see Table A2 in [Szerman \(2023\)](#) for more details about the firm population).

G.2 Survey Questions

The survey consists of five complementary parts.

First, the survey begins with questions on affirmative action support. Respondents are asked to rate, on a four-point scale (1 = “disagree,” 4 = “agree a lot”), how much they agree with the following statements: (i) “women should have preference in hiring in the labor market”; (ii) “Black people should have preference in hiring in the labor market”; (iii) “people with disabilities should have preference in hiring in the labor market”; and (iv) “hiring quotas for people with disabilities should exist.” The order of the questions, as well

as whether respondents see one or both of the last two items, is randomized

Second, the survey includes questions adapted from [Alfaro-Urena et al. \(2021\)](#). Respondents are asked to select up to three alternatives indicating: (i) “which are the most important aspects your company uses to select highly qualified workers (e.g., managers, engineers, and administrators)”; (ii) “which are the most important aspects your company uses to select less qualified workers (e.g. operators, packers, and janitorial staff)”; and (iii) “which are the most important aspects your company uses to select workers with disabilities”. The alternatives are: (a) curriculum vitae; (b) letters of recommendation or references from former employers or teachers; (c) immediate availability; (d) test of cognitive, psychometric, or psychological skills; (e) test of knowledge or professional skills related to the job; (f) evaluation during the probationary period; (g) interviews or tests using English or other languages; (h) criminal records; and (j) availability of infrastructure or assistive technologies. The question order is randomized.

Third, the survey includes a vignette experiment. Respondents are shown a scenario describing a large fictitious consulting company seeking to hire an entry-level employee to perform routine clerical and organizational tasks. The vignette introduces Rafael, a 22-year-old high school graduate with flexibility, proactivity, good organizational skills, and strong interpersonal abilities. For a random subset of respondents, the vignette additionally states that Rafael has bilateral hearing loss. This disability is chosen because it is very common and does not require accommodations or reduce productivity for this type of job. Respondents rate, on a four-point scale (1 = “unlikely,” 4 = “very likely”), how likely they think the company would be to hire him and how likely they think he would accept the job.

Fourth, the survey incorporates questions adapted from [Domzal et al. \(2008\)](#) about employer perceptions regarding the employment of people with disabilities. Respondents rate, on a four-point scale (1 = “disagree,” 4 = “agree a lot”), the extent to which each of the following issues represents a barrier to hiring workers with disabilities at their company: (i) lack of knowledge or information; (ii) accommodations or barriers during the hiring process; (iii) attitudes of co-workers; (iv) attitudes of supervisors; (v) fear of absenteeism; (vi) accommodation costs; (vii) difficulty finding qualified candidates with disabilities; and (viii) a belief that the nature of the work cannot be performed by people with disabilities. For item (vi), respondents are also asked to provide examples of accommodation costs incurred by the firm to verify whether their concerns align with actual expenses.

Next, respondents are presented with some concerns about people with disabilities often cited by employers. Using the same four-point scale, they indicate how much they agree that each concern is relevant at the company: (i) supervisors are not comfortable managing employees with disabilities; (ii) supervisors are unsure how to evaluate these employees; (iii) co-workers are uncomfortable working alongside employees with disabilities; (iv) workers with disabilities lack the necessary skills and experience; (v) workers with disabilities may be less safe or productive than others; (vi) hiring workers with disabilities increases management time; and (vii) negative stigma associated with disability hiring quotas.

The survey then asks about some strategies that companies may use to reduce barriers to hiring workers with disabilities. Respondents again rate, on a four-point scale, how useful they believe each strategy is: (i) using recruitment agencies specializing in candidates with disabilities; (ii) creating a diversity committee and training existing staff; (iii) increasing availability of assistive technologies; (iv) offering flexible working hours; (v) fiscal incentives, such as subsidies or tax deductions; and (vi) more labor inspections.

Fifth, the survey concludes with open-ended questions about the performance of the firm's most recent hire with disabilities relative to other employees, whether any accommodations were provided, and the respondent's perceptions of what triggered labor inspections. These questions allow respondents to describe, in their own words, experiences and mechanisms related to disability hiring at their workplace.

G.3 Survey Results

I discuss the survey results in two ways. First, I report the average means of the responses across control and treatment firms, which correspond to firms without and under AA quota requirements (i.e., firms with fewer and more than 100 employees). Second, I show differences in responses between these two groups, along with standard errors. These differences are robust to the inclusion of type of questionnaire fixed effects to account for the fact that the order of some questions is randomized.

Affirmative Action Support. Panel A of Table G2 displays results for affirmative action support. Relative to black people and women, respondents are, on average, more likely to support preference in hiring for people with disabilities. There are no statistically significant differences across groups of firms. The p-values range from 0.330 to 0.967.

Vignette Experiment. For the vignette experiment, I estimate the following specification:

$$R_{ij} = \alpha + \beta_1 \times \mathbf{1}(\text{Quota}_i \geq 100) \times \mathbf{1}(\text{Has Disability}_i = 1) + \beta_2 \times \mathbf{1}(\text{Quota}_i \geq 100) + \beta_3 \times \mathbf{1}(\text{Has Disability}_i = 1) + \gamma_j + \epsilon_{ij}, \quad (8)$$

in which R_{ij} stands for firm i 's answer in questionnaire type j ; $\mathbf{1}(\text{Quota}_i \geq 100)$ is an indicator for treated firms with at least 100 employees; $\mathbf{1}(\text{Has Disability}_i = 1)$ stands for firm i 's question receiving randomized information on disability; and γ_j is the questionnaire type fixed effects. Standard errors are clustered at the firm level.

Table G1 reveals results for the vignette experiment. Firms are, on average, 22.2 percent less likely to express interest in hiring a fictitious candidate with disabilities relative to a similar counterpart without disabilities. Firms do not display significant differences in the perceived likelihood that a fictitious candidate will accept the position.

Table G1: Survey: Vignette Experiment

	(1) Hire	(2) Accept
Above 100 emp.	-0.093 (0.193)	0.019 (0.169)
Has Disability	-0.723** (0.309)	-0.150 (0.301)
Has Disability x Above 100 emp.	0.403 (0.271)	0.291 (0.259)
Mean Dep. Var (Control)	3.258	3.194
Mean Dep. Var (Treated)	3.345	3.345

Note: Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the means of answers for firms with fewer and more than 100 employees, along with point estimates after estimating Equation (8). The final sample has 60 firms, with 31 firms having more than 100 employees.

Most Important Aspects to Select Workers. Table G2 reports the results on the most important aspects used to select highly skilled, less skilled, and disabled workers. Curriculum vitae (CV) is the most frequently cited factor for all three groups. Several additional patterns

emerge. First, the factors employers prioritize when hiring workers with disabilities more closely resemble those used for hiring less skilled workers than those used for recruiting highly skilled workers. Second, approximately 34 percent of employers report that the availability of infrastructure or assistive technologies is an important criterion, but this share does not differ significantly between quota and non-quota firms (p-value = 0.528). Third, firms subject to the AA quota are more likely to use the probationary period (p-value = 0.002) and less likely to use knowledge tests (p-value = 0.111) when selecting workers with disabilities. Fourth, although larger firms are more likely to rely on CVs when selecting both highly skilled (p-value = 0.054) and less skilled workers (p-value = 0.058), no such differential emerges for workers with disabilities (p-value = 0.304).

Employer Perceptions: Challenges, Concerns, and Reducing Barriers in Hiring People with Disabilities. Panel B of Table G3 reveals several challenges in hiring people with disabilities. I find that the most relevant challenges are the following: lack of knowledge or information about people with disabilities, inability to find qualified people with disabilities, and accommodations and barriers during the hiring process. These patterns seem consistent with employers having more challenges in interpreting signals from people with disabilities. As indirect evidence of lower hiring standards to recruit more workers with disabilities (Coate and Loury, 1993), I also find that firms under AA quota are more likely to report that they cannot find qualified workers with disabilities (p-value = 0.072). Rates for attitudes of co-workers and supervisors as challenges are lower. On the other hand, firms under AA quota are more likely to report the attitudes of co-workers (p-value = 0.000) and supervisors (p-value = 0.012) as challenges.

Panel C turns to concerns about workers with disabilities at the company. Supervisors who are not sure how to evaluate a person with a disability appear as the most relevant concern, followed by supervisors who are not comfortable managing workers with disabilities. Firms under the AA quota are also more likely to report that people with disabilities might not be as safe and productive as other workers without disabilities (p-value = 0.032), workers with disabilities require additional management time (p-value = 0.136), and co-workers are not comfortable with colleagues with disabilities (p-value = 0.039). These results are consistent with evidence that workers with disabilities have lower educational levels.

Lastly, Panel D illustrates the most relevant strategies to reduce barriers to hiring peo-

ple with disabilities. Respondents tend to disagree that more frequent labor inspections and flexible working hours would represent important strategies. Interestingly, there is no evidence that firms under the AA quota are more likely to report investments in screening capital (e.g., specialized recruitment agencies) and workplace accommodations (e.g., increasing availability of assistive technologies) as relevant strategies. If anything, firms under the AA quota are *less* likely to agree that specialized recruitment agencies as relevant strategies (p-value = 0.050).

Open-Ended Questions. Among other questions, the survey asks how likely respondents think on a four-point scale, in which 1 is “disagree” and 4 is “agree a lot”, that inspections are triggered by employees. I find that 41.67 percent of respondents report to be likely or very likely. When asked who they think made these complaints, employees with disabilities are never mentioned.

Table G2: Survey Responses: Affirmative Action Support and Employer Perceptions

	(1) Mean (Control)	(2) Mean (Treated)	(3) Diff. (1)-(2)	(4) (Std. Err.)	(5) P-Value
Panel A: Affirmative Action Support					
<i>Women should be given preference in hiring</i>	2.323	2.333	-0.011	(0.260)	0.967
<i>Blacks should be given preference in hiring</i>	2.194	2.231	-0.037	(0.252)	0.883
<i>PwD should be given preference in hiring</i>	3.062	2.714	0.348	(0.351)	0.330
<i>Hiring quotas for PwD should exist</i>	3.333	3.467	-0.133	(0.230)	0.566
Panel B: Challenges in Hiring People with Disabilities					
<i>Lack of knowledge or information about people with disabilities</i>	2.586	2.464	0.122	(0.258)	0.638
<i>Accommodations or barriers during the hiring process</i>	2.161	2.414	-0.253	(0.257)	0.329
<i>Attitudes of co-workers</i>	1.367	2.172	-0.806***	(0.210)	0.000
<i>Attitudes of supervisors</i>	1.400	2.000	-0.600**	(0.232)	0.012
<i>Fear of absenteeism</i>	1.452	1.759	-0.307	(0.201)	0.131
<i>Not knowing how much accommodation will cost</i>	1.613	1.621	-0.008	(0.215)	0.971
<i>Cannot find qualified PwD</i>	2.226	2.690	-0.464*	(0.253)	0.072
<i>Nature of the work cannot be performed by PwD</i>	1.677	1.759	-0.081	(0.225)	0.720
Panel C: Concerns for the Company					
<i>Supervisors are not comfortable managing PwD</i>	1.700	2.000	-0.300	(0.224)	0.186
<i>Supervisors are not sure how to evaluate a PwD</i>	2.034	2.034	0.000	(0.253)	1.000
<i>Co-workers are not comfortable with colleagues with disabilities</i>	1.333	1.759	-0.425**	(0.201)	0.039
<i>PwD lack the skills and experience to do their jobs</i>	1.355	1.552	-0.197	(0.204)	0.339
<i>PwD may not be as safe and productive as other workers</i>	1.484	2.000	-0.516**	(0.234)	0.032
<i>It costs more due to additional management time</i>	1.567	1.897	-0.330	(0.218)	0.136
<i>Negative stigma associated with hiring quota for PwD</i>	1.742	1.759	-0.017	(0.228)	0.942
Panel D: Strategies to Reduce Barriers in Hiring People with Disabilities					
<i>Specialized recruitment agencies</i>	3.065	2.621	0.444**	(0.222)	0.050
<i>Diversity committee and training existing staff</i>	2.839	2.828	0.011	(0.204)	0.957
<i>Increasing availability of assistive technologies</i>	2.839	2.750	0.089	(0.187)	0.638
<i>Flexible working hours</i>	2.323	2.586	-0.264	(0.235)	0.266
<i>Fiscal incentives, such as subsidies and tax deductions</i>	2.677	2.793	-0.116	(0.244)	0.637
<i>More labor inspections</i>	2.323	2.276	0.047	(0.255)	0.855

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the means of answers for firms with fewer and more than 100 employees, along with differences in means, standard errors, and p-values. The final sample has 60 firms, with 31 firms having more than 100 employees.

Table G3: Survey Responses: Affirmative Action Support and Employer Perceptions

	(1) Mean (Control)	(2) Mean (Treated)	(3) Diff. (1)-(2)	(4) (Std. Err.)	(5) P-Value
Panel A: Three Most Important Factors to Select People With Disabilities					
<i>Curriculum Vitae</i>	0.633	0.759	-0.125	(0.121)	0.304
<i>Letters of recommendations or references from former employers/teachers</i>	0.333	0.207	0.126	(0.117)	0.283
<i>Immediate availability</i>	0.100	0.207	-0.107	(0.094)	0.261
<i>Test of cognitive, psychometric, or psychological skills</i>	0.167	0.103	0.063	(0.090)	0.487
<i>Test of knowledge or professional skills related to the job</i>	0.400	0.207	0.193	(0.119)	0.111
<i>Evaluation during the probationary period</i>	0.367	0.759	-0.392***	(0.121)	0.002
<i>Interview or test using English or other languages</i>	0.167	0.172	-0.006	(0.099)	0.954
<i>(Lack of) Criminal records</i>	0.100	0.069	0.031	(0.074)	0.675
<i>Availability of infrastructure or assistive technologies</i>	0.300	0.379	-0.079	(0.125)	0.528
Panel B: Three Most Important Factors to Select Less Skilled Workers					
<i>Curriculum Vitae</i>	0.742	0.929	-0.187*	(0.096)	0.058
<i>Letters of recommendations or references from former employers/teachers</i>	0.484	0.250	0.234*	(0.124)	0.065
<i>Immediate availability</i>	0.387	0.321	0.066	(0.127)	0.606
<i>Test of cognitive, psychometric, or psychological skills</i>	0.161	0.143	0.018	(0.095)	0.847
<i>Test of knowledge or professional skills related to the job</i>	0.226	0.214	0.012	(0.110)	0.917
<i>Evaluation during the probationary period</i>	0.387	0.571	-0.184	(0.130)	0.162
<i>Interview or test using English or other languages</i>	0.161	0.179	-0.017	(0.099)	0.863
<i>(Lack of) Criminal records</i>	0.097	0.143	-0.046	(0.086)	0.592
Panel C: Three Most Important Factors to Select Highly Skilled Workers					
<i>Curriculum Vitae</i>	0.645	0.862	-0.217*	(0.110)	0.054
<i>Letters of recommendations or references from former employers/teachers</i>	0.419	0.552	-0.132	(0.130)	0.313
<i>Immediate availability</i>	0.129	0.172	-0.043	(0.094)	0.645
<i>Test of cognitive, psychometric, or psychological skills</i>	0.323	0.379	-0.057	(0.125)	0.652
<i>Test of knowledge or professional skills related to the job</i>	0.484	0.448	0.036	(0.131)	0.787
<i>Evaluation during the probationary period</i>	0.484	0.276	0.208	(0.125)	0.101
<i>Interview or test using English or other languages</i>	0.161	0.207	-0.046	(0.101)	0.655
<i>(Lack of) Criminal records</i>	0.129	0.069	0.060	(0.078)	0.447

Note: ***: significant at 1% level; **: significant at 5% level; *: significant at 10% level. This table reports the means of answers for firms with fewer and more than 100 employees, along with differences in means, standard errors, and p-values. The final sample has 60 firms, with 31 firms having more than 100 employees.

H Welfare Framework

I propose a framework to guide the welfare analysis of enforcing AA quotas. The model is parsimonious and delivers the minimal structure to map firms' compliance decisions under inspections into changes in employment, producer surplus, and fiscal revenues.

Model Setup. Consider a one-period economy with a unit mass of firms and workers with disabilities. Each firm faces a *firm-specific cost* σ of hiring a worker with disabilities such that:

$$\sigma \sim F(\sigma), \quad f(\sigma) > 0,$$

in which $F(\cdot)$ is the distribution function and $f(\cdot)$ its density. The cost σ captures all real and perceived costs associated with hiring a worker with disabilities, including accommodation, supervision, administrative burden, perceived productivity differences, or taste-based costs.

If a firm hires a worker with disabilities, it gets the surplus from that hire, defined as $MRPL_d - w - \sigma$, in which $MRPL_d$ is the marginal revenue product generated by the worker with disabilities and w is the wage paid to this worker.³⁴

Firms are subject to AA quota regulation but may choose not to comply initially. Let σ^* denote the *enforcement level* chosen by the government (e.g., inspection frequency). Conditional on being non-compliant, a firm is detected with probability:

$$p(\sigma^*), \quad p'(\sigma^*) > 0,$$

so that stricter enforcement mechanically raises the likelihood of detection. Importantly, the firm-specific cost σ is a primitive characteristic of the firm and conceptually distinct from the enforcement level σ^* , which is a policy variable chosen by the government that affects only the probability of detecting non-compliance, not the firm's underlying cost. While the government does not observe σ , it can adjust σ^* to raise the average probability that non-compliance is detected.

If detected, a non-compliant firm chooses between complying with the quota or paying the statutory fine. Compliance after detection yields $M\tilde{R}PL_d - w - \sigma$, in which $M\tilde{R}PL_d$ allows for additional costs triggered by detection, such as bureaucratic hassle or follow-up

³⁴For simplicity, I take wages as given. This is consistent with institutional wage rigidity and with the fact that workers with disabilities are predominantly employed in low-skill, low-wage jobs.

checks. Paying the fine yields $-F$. If a non-compliant firm is not detected, it hires no worker with disabilities and its surplus on this margin is zero.

Lastly, an employed person with disabilities receives utility $u(w - \tau)$, where w is the wage and τ is a lump-sum tax. An unemployed person with disabilities receives a welfare benefit payment b and utility $v(b)$. The functions $u(\cdot)$ and $v(\cdot)$ are increasing and concave.

Compliance Thresholds. Define $\sigma^F = M\tilde{R}PL_d + F - w$ as the cutoff value at which a detected firm is indifferent between complying and paying the fine. Firms with $\sigma \leq \sigma^F$ comply upon detection, and those with $\sigma > \sigma^F$ pay the fine. Similarly, define σ^C as the cutoff value that makes a firm indifferent between complying immediately and delaying compliance:

$$MRPL_d - w - \sigma^C = p(\sigma^*)(M\tilde{R}PL_d - w - \sigma^C) \Rightarrow \sigma^C = \frac{(MRPL_d - w) - p(\sigma^*)(M\tilde{R}PL_d - w)}{1 - p(\sigma^*)}.$$

Under standard assumptions, it follows that $\sigma^C < \sigma^F$. Thus, there is a positive mass of firms that comply immediately ($\sigma \leq \sigma^C$), a mass that comply only if detected ($\sigma^C < \sigma \leq \sigma^F$), and a mass that always pay the fine ($\sigma > \sigma^F$).

Decision Trees. Figures H1 and H2 illustrate how the interaction between government inspections and firms' compliance decisions determines payoffs for both firms and individuals.

Figure H1: Decision Tree with Firm Payoffs

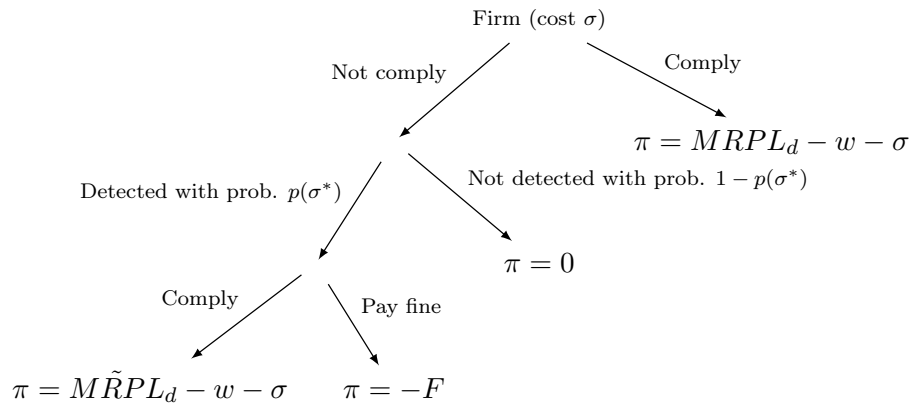
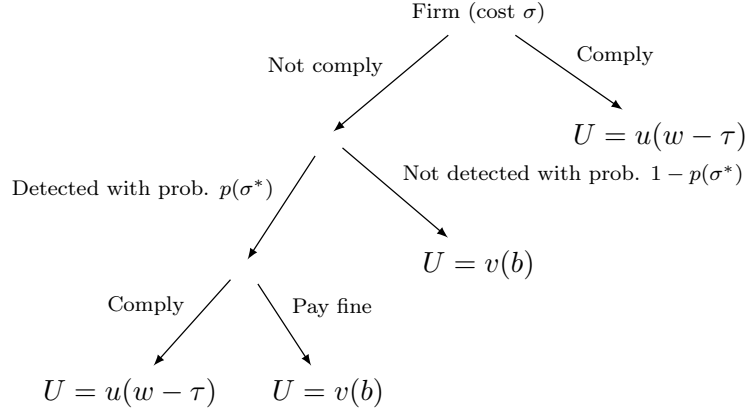


Figure H2: Decision Tree with Individual Payoffs



Firms. Given w , the producer surplus associated with hiring workers with disabilities is:

$$\begin{aligned} \Pi(\sigma^*) = \int_0^{\sigma^C} [MRPL_d - w - \sigma] f(\sigma) d\sigma + \int_{\sigma^C}^{\sigma^F} p(\sigma^*) [M\tilde{R}PL_d - w - \sigma] f(\sigma) d\sigma \\ - \int_{\sigma^F}^1 p(\sigma^*) F f(\sigma) d\sigma. \end{aligned} \quad (9)$$

in which the first term corresponds to firms that comply with the AA quota from the beginning; the second term to firms that comply only after detection; and the third term to firms that pay the fine when detected.

People with Disabilities. Given w , τ , and b , total welfare for people with disabilities is:

$$\begin{aligned} V(\sigma^*) = \int_0^{\sigma^C} u(w - \tau) f(\sigma) d\sigma + \int_{\sigma^C}^{\sigma^F} p(\sigma^*) u(w - \tau) f(\sigma) d\sigma \\ + \int_{\sigma^F}^1 p(\sigma^*) v(b) f(\sigma) d\sigma + \int_{\sigma^C}^1 [1 - p(\sigma^*)] v(b) f(\sigma) d\sigma, \end{aligned} \quad (10)$$

in which the terms correspond, respectively, to employment due to initial compliance; employment due to compliance after detection; benefit receipt when fines are paid after detection; and benefit receipt when non-compliance is not detected.

Government. Fiscal revenues consist of taxes from employed workers with disabilities and fines from detected non-compliers, net of welfare benefits paid to unemployed workers. Total

revenues can be written as:

$$R(\sigma^*) = \tau \int_0^{\sigma^C} f(\sigma) d\sigma + \tau \int_{\sigma^C}^{\sigma^F} p(\sigma^*) f(\sigma) d\sigma + F \int_{\sigma^F}^1 p(\sigma^*) f(\sigma) d\sigma - b \int_{\sigma^F}^1 p(\sigma^*) f(\sigma) d\sigma - b \int_{\sigma^C}^1 [1 - p(\sigma^*)] f(\sigma) d\sigma. \quad (11)$$

The government also incurs an enforcement cost $C(\sigma^*)$, which can reflect the administrative cost of inspections and related activities.

Marginal Effects of Enforcement of AA Quotas. Total social welfare is defined as:

$$W(\sigma^*) = \Pi(\sigma^*) + V(\sigma^*) + R(\sigma^*) - C(\sigma^*). \quad (12)$$

In addition, for convenience, I define $M_C \equiv \int_{\sigma^C}^{\sigma^F} p'(\sigma^*) f(\sigma) d\sigma$, which captures the mechanical increase in the measure of firms that comply upon detection when enforcement becomes more stringent, and $M_F \equiv \int_{\sigma^F}^1 p'(\sigma^*) f(\sigma) d\sigma$, which represents the marginal change in the mass of firms that pay fines. With a slight abuse of notation, I interpret $M\tilde{R}PL_d$ as the marginal revenue product of workers with disabilities net of the expected firm-specific cost σ among the firms that are marginal with respect to enforcement.³⁵

The welfare impact for people with disabilities of a small change in enforcement can be written as:

$$\frac{\partial V}{\partial \sigma^*} = M_C[u(w - \tau) - v(b)], \quad (13)$$

in which M_C represents the mass of inframarginal firms that switch from delinquency to compliance due to a small change in enforcement.³⁶

Analogously, the impact on firm profits is:

$$\frac{\partial \Pi}{\partial \sigma^*} = (M\tilde{R}PL_d - w)M_C - M_FF, \quad (14)$$

which incorporates the change in surplus from additional hires by compliant firms and from

³⁵This term absorbs $\int_{\sigma^C}^{\sigma^F} \sigma f(\sigma) d\sigma$ into $M\tilde{R}PL_d$, allowing the marginal change in producer surplus to be written as $(M\tilde{R}PL_d - w)M_C - M_FF$. This keeps the welfare expressions simple and does not affect the subsequent welfare conclusions.

³⁶The policy has no welfare effect for firms that always abstain from compliance regardless of enforcement level. In addition, by the envelope theorem, there is no welfare effect from firms that always comply with the regulation regardless of enforcement level.

finer paid by non-compliant firms.

Finally, the effect on government revenues can be expressed as:

$$\frac{\partial R}{\partial \sigma^*} = M_C[\tau + b] + M_FF, \quad (15)$$

in which the marginal fiscal gain reflects higher tax revenues and lower benefit payments from new hires, plus additional fine revenues.

Combining Equations (13)–(15), the derivative of social welfare with respect to enforcement can be represented as:

$$W'(\sigma^*) = M_C[MR\tilde{P}L_d - w] + M_C[u(w - \tau) - v(b)] + M_C[\tau + b] - C'(\sigma^*). \quad (16)$$

Equation (16) highlights four objects that govern the welfare effects of stronger enforcement: (i) the change in producer surplus, which depends on the wedge between the marginal revenue product of workers with disabilities and their wages; (ii) the change in surplus for people with disabilities from working relative to receiving welfare benefit payments; (iii) the fiscal gain from higher tax revenues and lower welfare benefit payments; and (iv) the marginal cost of enforcement. Because the empirical results show that workers without disabilities are unaffected in terms of wages and employment, there is no welfare change for them in this framework. By contrast, raising fiscal revenues through increased employment can finance additional public goods that benefit both people with and without disabilities.

Incumbent Workers with Disabilities. The model does not distinguish between new hires and incumbent workers with disabilities and assumes fixed wages. If firms adjust through lower wage growth for incumbent workers with disabilities, such adjustment represents a transfer from people with disabilities to firms under linear utility. This implies that the government only needs to trade off marginal benefits and marginal producer loss from new hires with disabilities when evaluating total welfare. Similarly, revenues from fines represent a transfer from firms to the government under linear utility.

Optimal Enforcement. The government chooses the enforcement level σ^* to maximize $W(\sigma^*)$. Under standard regularity conditions, the first-order condition can be written as:

$$\underbrace{M_C[u(w - \tau) - v(b)]}_{\text{marginal welfare benefit}} + \underbrace{M_C[\tau + b]}_{\text{marginal revenue benefit}} = \underbrace{C'(\sigma^*)}_{\text{marginal cost of enforcement}} + \underbrace{M_C(w - \tilde{MRPL}_d)}_{\text{marginal producer cost}}. \quad (17)$$

Calibration: Marginal Effects of Enforcement of AA Quotas. All relevant objects can be related to quantities observed in the data or inferred from reduced-form estimates. The marginal welfare benefit depends on the job surplus; the marginal revenue benefit can be obtained from administrative data; the marginal cost of enforcement can be bounded by the average cost of inspections; and the marginal producer loss can be recovered using the discrete choice framework discussed in Section 6.

Marginal Welfare Benefit. To overcome the lack of data on consumption and assets and to obtain a tractable expression, I assume hand-to-mouth agents so that consumption tracks net income. This assumption is reasonable in this context. I also focus on the flow of income, implying that my estimates are conservative and reflect a lower bound of the benefit side.

I take two approaches to derive bounds on the marginal welfare benefit. First, to obtain an upper bound, I assume that unemployed individuals do not have a marginal disutility from working. In this case, I approximate the marginal benefit as the flow of income from switching from welfare benefits to employment. From the 2010 Census, I calculate that, on average, a worker with disabilities in the formal sector earns 22,660 Brazilian *reais* per year and is subject to an income tax rate of 7.5 percent. Each person with disabilities not employed in the formal sector receives 3,827.16 Brazilian *reais* in welfare benefits. Given the 50 percent increase in the number of employees, together with a baseline job rate of 0.31, the marginal welfare benefit associated with additional hires is 2,655.67 Brazilian *reais*.

Second, to obtain a lower bound, I incorporate the opportunity cost of a full-time job due to lost leisure. I follow [Mas and Pallais \(2019\)](#) and assume that the value of non-work time relative to labor earnings is 0.58. Under this assumption, the estimated marginal welfare benefit is 771.32 Brazilian *reais*.

Marginal Revenue Benefit. The marginal fiscal gain is the sum of revenues raised through income tax and saved on welfare benefits for new hires. The marginal revenue benefit is

856.63 Brazilian *reais*.

Marginal Cost of Enforcement. Because there is no data available on inspection-specific spending, I use the average cost of inspections as an upper bound for the marginal cost of enforcement. According to the Federal Budget Panel (*Painel do Orçamento Federal*), the average annual budget for labor inspection (deflated to 2018 prices) is 27,787,724 Brazilian *reais*. Considering that the total number of inspections per year is about 279,857, each inspection costs around 99.29 Brazilian *reais*. Other enforcement policies that can boost compliance and target non-compliant firms more directly, such as data-driven inspections and threat-of-audit letters, are possible alternatives. While these policies are likely to reduce the costs of inspections in the long term, implementing them would also be costly.

Marginal Producer Cost. Due to the lack of data on profits, changes in producer surplus cannot be directly inferred. Instead, I rely on two indirect approaches to construct bounds for the marginal producer loss. First, I propose a discrete choice framework based on firms’ decisions to comply with the AA quotas, discussed in detail in Section 6.

Second, I use reduced-form estimates to back out a “break even” producer cost that sets marginal welfare benefits equal to marginal costs. Table H1 reports the “break even” values of the average ratio between the marginal revenue product of labor and average wages considering different scenarios. Each row indicates whether the opportunity cost of work and fiscal revenue benefit enter into the calculation. Even in the most conservative scenario, the ratio is below one, indicating that social benefits exceed social costs.

Table H1: Cost-Benefit Analysis

	(1)	(2)	(3)
	Opport. Cost of Work	Revenue Benefit	Break-Even MRPL/w
No		Yes	0.0283
Yes		Yes	0.5647
No		No	0.2721
Yes		No	0.8086

Note: This table reports a range of “break-even” values of $\tilde{MRPL}_d$, normalized by average wages, that set marginal welfare benefits equal to marginal costs of increasing enforcement of AA quotas in Equation (16).

Discussion and Limitations. This framework abstracts from several channels that may generate additional social benefits or costs. On the benefit side, I do not account for potential improvements in health and longevity from employment, crowd out from public to private health care, or lower criminal involvement associated with higher employment among people with disabilities (Paim et al., 2011; Sullivan and Von Wachter, 2009; Deshpande and Mueller-Smith, 2022). On the cost side, I do not model welfare losses for workers without disabilities, distortions in the production function, or moral hazard effects. Because the reduced-form analysis shows little evidence that enforcement of AA quotas affects workers without disabilities or firm outcomes, the social benefits from enforcement are likely to be underestimated in this framework.

Another limitation is that the social welfare function implicitly assigns equal social weights to firms and people with disabilities. The prevalence of quota policies worldwide suggests that governments may place higher social value on improving the welfare of people with disabilities. Estimating these welfare weights is beyond the scope of this paper and is an important area for future research.